Fundamentals of Deep Learning

Lecture Notes

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November 4, 2026

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Chapter 1

From Logistic Regression to MLPs

1.1 Recap of Logistic Regression

The question behind every formula

A classifier must decide what to do, but first it can estimate what might be true. Logistic regression separates these two tasks: a model assigns a probability to a binary outcome, and a decision rule turns that probability into an action. The sigmoid specifies the probability model; cross-entropy measures how much probability that model assigned to the outcome actually observed.

1.1.1 The problem: a decision is not a probability

Imagine a robot about to grasp a cup. From its camera and estimated gripper pose, it measures a centering error and an angular misalignment. The outcome is $Y = 1$ for a successful grasp and $Y = 0$ for a failure. We would like to answer both "How likely is success?" and "Should the robot attempt this grasp?" These are different questions. A success probability of 0.7 might justify a cheap retryable grasp but not a grasp whose failure breaks an expensive object.

Why not just fit a line to the labels? A real-valued affine prediction can be negative or greater than one. Why not immediately threshold that line? A hard threshold throws away the distinction between a barely positive score and a strongly positive score. It also gives a piecewise constant classification error, which offers little useful gradient information. We therefore seek a smooth conditional probability model before choosing a decision rule.

This section develops the connected chain

$$\text{features} \longrightarrow \text{score} \longrightarrow \text{probability} \longrightarrow \text{likelihood} \longrightarrow \text{loss} \longrightarrow \text{learning and decisions.}$$

We first understand a fixed model. Only later do we vary its parameters to train it. The examples and plots here are constructed teaching examples, not empirical claims about a deployed robot.

1.1.2 Notation and the assumptions we are making

Let $x = (x_1, \ldots, x_d)^\top \in \mathbb{R}^d$ denote the measured features and let $Y \in \{0, 1\}$ be a random outcome. Define the augmented input and parameter vector

$$\tilde{x} = \begin{bmatrix} 1 \\ x \end{bmatrix} \in \mathbb{R}^{d+1}, \qquad w = \begin{bmatrix} b \\ a \end{bmatrix}, \qquad z = w^\top \tilde{x} = b + a^\top x.$$

The slides call the bias w_0 and use $x_0 = 1$; thus their $w^\top x$ is our $w^\top \tilde{x}$. We keep the tilde to distinguish the actual feature space from the augmented space. The scalar z is the *score* or *logit*.

Symbol	Meaning
Y	Random binary label before observing the outcome
y	Observed label, exactly zero or one
$q(x)$	True conditional probability $\Pr(Y = 1 \mid X = x)$
$p_w(x)$	Model probability, also written $h_w(x)$
$\hat{y}$	In the slides, the probability $h_w(x)$
$\hat{c}$	Hard class prediction, zero or one
$\ell(p, y)$	Loss on one observation
$J(w)$	Mean loss over the training data

Definitions, assumptions, consequences. That a binary conditional distribution is Bernoulli is a consequence of its two possible outcomes: it is completely specified by one probability. That its *log-odds is affine in the chosen features* will be our modeling assumption. That independent observations give a product likelihood is an additional sampling assumption. Cross-entropy then follows from maximum likelihood; its convexity follows from calculus. These statements play different logical roles.

Conditioning matters. We model $Y \mid X = x$, not one identical Bernoulli law for all inputs. At different inputs the success probabilities can differ. Treating Y probabilistically does not assert that the world is fundamentally random. Unmeasured friction, occlusion, noisy sensors, or missing features can make an outcome unpredictable from the information we have.

1.1.3 Inventing the sigmoid through odds and evidence

A useful probability must stay in $[0, 1]$, while a linear score can take any real value. Many maps could enforce this range. Why use this particular one?

For $0 < p < 1$, define the *odds* as $p/(1 - p)$. If $p = 0.8$, the odds are 4: success has four times the probability of failure. Odds live in $(0, \infty)$; logarithms turn this into the whole real line. The *log-odds* is

$$\operatorname{logit}(p) = \log \frac{p}{1 - p}.$$

We now *assume* that evidence measured in log-odds is additive:

$$\log \frac{p_w(x)}{1 - p_w(x)} = b + \sum_{j=1}^{d} a_j x_j = z. \tag{1.1}$$

Solving, rather than memorizing, gives

$$\frac{p}{1-p} = e^z \quad \Longrightarrow \quad p = e^z(1-p) \quad \Longrightarrow \quad p(1+e^z) = e^z \quad \Longrightarrow \quad \boxed{p_w(x) = \sigma(z) = \frac{1}{1+e^{-z}}.}$$

This is an equivalence: affine log-odds and a sigmoid of an affine score describe exactly the same model. Binary labels alone do not force this choice; a probit link, for example, would define a different model.

Equation (1.1) also explains the parameters. Increasing x_j by one unit with other features held fixed multiplies the odds by e^{a_j}. It does *not* add a_j to the probability. The bias is the log-odds at $x = 0$, so $p_w(0) = \sigma(b)$. This interpretation depends on units and on whether zero is meaningful for the features.

Why additive log-odds is a reasonable idea. Bayes' rule gives, wherever the densities are positive,

$$\log \frac{\Pr(Y=1 \mid x)}{\Pr(Y=0 \mid x)} = \log \frac{\Pr(Y=1)}{\Pr(Y=0)} + \log \frac{p(x \mid Y=1)}{p(x \mid Y=0)}.$$

Under conditional independence of features, the last term becomes a sum of featurewise log-likelihood ratios. This motivates adding evidence. Logistic regression itself does *not* require conditional feature independence; it models the final conditional log-odds directly. This is a conceptual motivation, not a claim about the historical invention of the method.

A nontrivial case where the affine assumption is exact. Suppose $X \mid Y = k \sim \mathcal{N}(\mu_k, \Sigma)$, with the same positive-definite covariance for both classes and priors $\pi_k > 0$. The normalizing constants cancel in the density ratio, leaving

$$\begin{aligned}\log \frac{q(x)}{1-q(x)} &= \log \frac{\pi_1}{\pi_0} - \frac{1}{2}(x-\mu_1)^\top \Sigma^{-1}(x-\mu_1) + \frac{1}{2}(x-\mu_0)^\top \Sigma^{-1}(x-\mu_0) \\ &= (\mu_1-\mu_0)^\top \Sigma^{-1} x - \frac{1}{2}(\mu_1^\top \Sigma^{-1}\mu_1 - \mu_0^\top \Sigma^{-1}\mu_0) + \log \frac{\pi_1}{\pi_0}.\end{aligned}$$

The quadratic terms in x cancel. Thus a Gaussian generative assumption can produce precisely a logistic conditional model. Unequal covariances generally leave quadratic terms. Logistic regression need not fit either Gaussian density to learn an affine log-odds model.

1.1.4 Geometry: why the boundary is a line

For equal costs of the two mistakes, predict class one when $p_w(x) \geq 1/2$. To understand its geometry, differentiate:

$$\sigma'(z) = \frac{e^{-z}}{(1+e^{-z})^2} = \sigma(z)(1-\sigma(z)) > 0 \quad \text{for finite } z.$$

The sigmoid is strictly increasing and $\sigma(0) = 1/2$. Therefore

$$\hat{c} = \mathbf{1}\{p_w(x) \geq 1/2\} = \mathbf{1}\{b + a^\top x \geq 0\}. \tag{1.2}$$

Its decision boundary is $b + a^\top x = 0$. In two dimensions this is a line; in three, a plane; in d dimensions, a hyperplane, provided $a \neq 0$. Strictly, $b + a^\top x$ is *affine* in x and linear in $\tilde{x}$. The informal phrase "linear classifier" includes an affine bias.

For the slide's $w = (-3, 1, 1)^\top$, we obtain

$$z = -3 + x_1 + x_2, \qquad x_2 = 3 - x_1.$$

At $(1,1)$ the score is -1 and $p \approx 0.269$; at $(1,2)$ it is 0 and $p = 0.5$; at $(2,2)$ it is 1 and $p \approx 0.731$; at $(4,4)$ it is 5 and $p \approx 0.9933$. We use class one for ties by convention. Another tie convention changes only points exactly on the boundary.

Direction, distance, and confidence. Moving by a displacement v changes the score by $a^\top v$. If $a^\top v = 0$, we remain on a constant-score hyperplane. Thus a is normal to the boundary. For $a \neq 0$ the signed perpendicular distance from x to the boundary is

$$\delta(x) = \frac{b + a^\top x}{\|a\|_2}, \qquad p_w(x) = \sigma(\|a\|_2 \delta(x)).$$

The score is a *scaled* distance, not a distance in physical units. Scaling all weights by $c > 0$ preserves the boundary but changes the probabilities to $\sigma(cz)$. As c grows, predictions away from the boundary approach zero or one. The same classification accuracy can hide very different probability quality. If $c < 0$, the boundary stays but class assignments reverse.

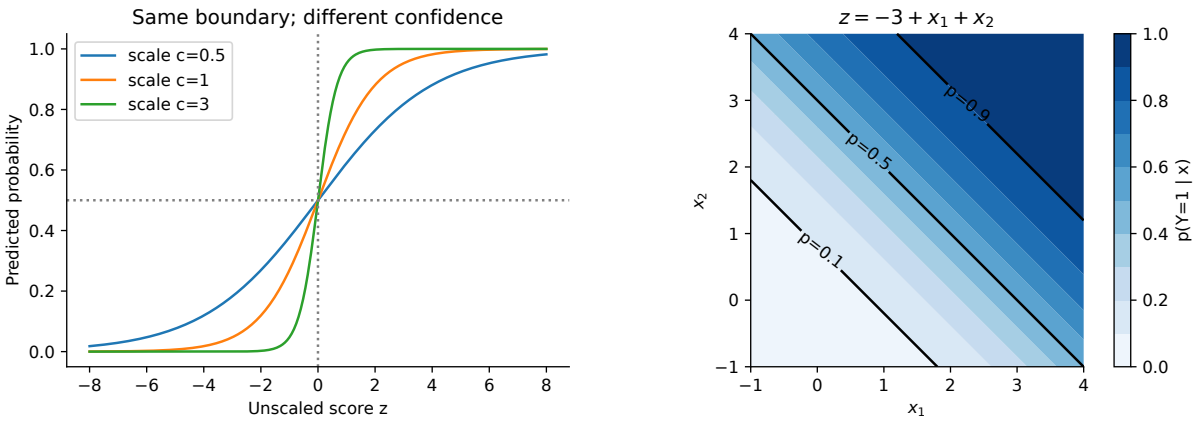


Figure 1.1: Left: rescaling a logit changes confidence without moving the zero threshold. Right: parallel probability contours around the boundary $x_1 + x_2 = 3$. The sigmoid bends the probability response, not this boundary.

Edge cases. If $a = 0$ and $b \neq 0$, the prediction is constant and there is no boundary. If $a = 0$ and $b = 0$, every point has probability $1/2$. The normalized-distance formula cannot be used in either case.

1.1.5 What does "maximally uncertain" actually mean?

For a fixed model w, $p_w(x)$ is a deterministic number at a fixed input. It specifies a distribution over Y, not a random choice of weights. The model's Bernoulli variance is

$$\mathrm{Var}(Y \mid x; w) = p(1-p) = \frac{1}{4} - (p - \tfrac{1}{2})^2,$$

which is largest at $p = 1/2$. Its predictive entropy, in nats, is

$$H(p) = -p \log p - (1-p) \log(1-p).$$

Differentiation gives

$$H'(p) = \log \frac{1-p}{p}, \qquad H''(p) = -\frac{1}{p} - \frac{1}{1-p} < 0.$$

So entropy has a unique maximum $H(1/2) = \log 2$. This is the precise sense in which the model is maximally uncertain at its 0.5 boundary.

Caution

A fitted logistic probability is not a guarantee of correctness. A probability near one may be unjustified on inputs unlike the training data. Predictive Bernoulli entropy does not separately measure uncertainty about the fitted parameters, missing model structure, or distribution shift. A decision boundary at a cost-dependent threshold other than 0.5 is not a maximum-entropy contour.

1.1.6 Nonlinear boundaries without nonlinear parameters

Replace raw features with a fixed map $\phi(x)$ and set $p_w(x) = \sigma(b + a^\top \phi(x))$. The boundary is affine in feature space, but can be curved in the original coordinates. For example,

$$\phi(x) = \begin{bmatrix} x_1^2 \\ x_2^2 \end{bmatrix}, \qquad z = 4(x_1^2 + x_2^2 - 1).$$

The 0.5 boundary is the circle $x_1^2 + x_2^2 = 1$; outside it the model predicts class one. In (ϕ_1, ϕ_2) coordinates it is the line $\phi_1 + \phi_2 = 1$. This is logistic regression with nonlinear basis functions. The score remains linear in its trainable coefficients.

A second example is XOR on $x_1, x_2 \in \{-1, 1\}$, with class one for opposite signs. A feature $x_1 x_2$ allows the score $z = -kx_1x_2$, $k > 0$, which separates all four points. An affine score in x_1, x_2 alone cannot: the midpoint of the two class-one points and the midpoint of the two class-zero points are both the origin, contradicting strict separation by an affine function.

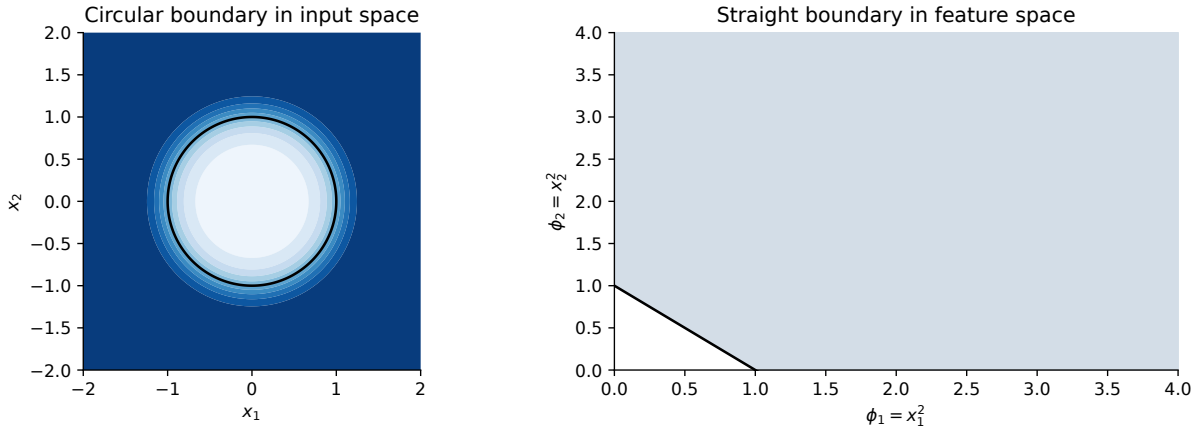


Figure 1.2: One model viewed in two coordinate systems. The curved boundary in input space becomes straight after the fixed feature transformation. Only the nonnegative quadrant is reachable in this feature space.

1.1.7 From a probability model to the likelihood of an observation

Once an input x is fixed, a Bernoulli model assigns

$$p_w(Y = y \mid x) = \begin{cases} p_w(x), & y = 1, \\ 1 - p_w(x), & y = 0. \end{cases}$$

Because y is binary, the two cases can be encoded as

$$p_w(Y = y \mid x) = p_w(x)^y \big(1 - p_w(x)\big)^{1-y}. \tag{1.3}$$

Check both values: when $y = 1$, the second exponent is zero; when $y = 0$, the first is zero. The formula does not make the label continuous. It uses the label as a selector of the probability assigned to the observed event.

Probability versus likelihood. With w fixed, this expression is a probability distribution over y; its two values sum to one. After observing (x, y), the very same expression viewed as a function of w is a *likelihood*. It asks which parameter choices would make the observed outcome plausible. It is not a probability distribution over w and need not integrate to one over parameter space. A Bayesian posterior over w would additionally require a prior and normalization.

For example, if a grasp succeeds, a model predicting $p = 0.9$ assigns likelihood 0.9, while one predicting $p = 0.2$ assigns 0.2. If that grasp fails, the likelihoods reverse their preference: 0.1 versus 0.8. A model is judged by what it said about the event that happened, not just by the thresholded class.

1.1.8 Deriving the training objective from maximum likelihood

Let $\mathcal{D} = \{(x_n, y_n)\}_{n=1}^N$ and $p_n = p_w(x_n)$. Assume the labels are conditionally independent across observations given their inputs and the common parameter w. An i.i.d. sample of pairs is a familiar sufficient setup. Then

$$\mathcal{L}(w; \mathcal{D}) = \prod_{n=1}^{N} p_n^{y_n} (1 - p_n)^{1-y_n}.$$

The product is a joint conditional probability of the observed labels. Adjacent video frames or repeated trials on the same object can violate the independence assumption. The sum loss can still be used as an objective in that case, but the simple independent-data likelihood interpretation and effective sample size need care.

We maximize this likelihood. Since log is strictly increasing, it preserves which positive value is largest. Since $\log(ab) = \log a + \log b$, it converts products into sums:

$$\begin{aligned}
\arg\max_w \mathcal{L}(w; \mathcal{D}) &= \arg\max_w \log \mathcal{L}(w; \mathcal{D}) \\
&= \arg\max_w \sum_{n=1}^{N} [y_n \log p_n + (1 - y_n) \log(1 - p_n)] \\
&= \arg\min_w \left\{ -\frac{1}{N} \sum_{n=1}^{N} [y_n \log p_n + (1 - y_n) \log(1 - p_n)] \right\}.
\end{aligned}$$

Negation turns maximization into minimization. Dividing by positive N does not change the minimizers; it expresses an average per observation. It does change gradient magnitudes, so the same numerical learning rate need not behave the same for sum and mean losses.

Definition 1.1 (Binary cross-entropy)**.** For one observation,

$$\boxed{\ell(p, y) = -y \log p - (1 - y) \log(1 - p).} \tag{1.4}$$

For a dataset, the mean negative log-likelihood is

$$\boxed{J(w) = \frac{1}{N} \sum_{n=1}^{N} \ell(p_w(x_n), y_n).} \tag{1.5}$$

All logarithms in this section are natural, so losses are measured in nats.

The brackets in (1.5) matter: the minus sign applies to both label-dependent log terms. Changing the logarithm base only multiplies the objective by a positive constant, leaving its minimizers unchanged.

1.1.9 Why a logarithmic penalty makes sense

The quantity $-\log r$ measures the *surprise* of an event to which the model assigned probability r. An event assigned probability one produces no surprise; an event assigned a tiny probability produces a large surprise. For independent events, surprise adds because their probabilities multiply. This is why the negative logarithm is so natural for a joint probability objective.

Probability of observed outcome r	0.9	0.5	0.01
Surprise $-\log r$	0.1054	0.6931	4.6052

For $y = 1$, $\ell = -\log p$ decreases as p increases, and $\partial \ell / \partial p = -1/p$. For $y = 0$, $\ell = -\log(1 - p)$ increases with p, and $\partial \ell / \partial p = 1/(1 - p)$. A confidently wrong prediction is much more expensive than an uncertain one. Correct classification alone does not make the loss zero: $p = 0.51$ and $p = 0.99$ both predict one, but their losses for $y = 1$ are about 0.673 and 0.010.

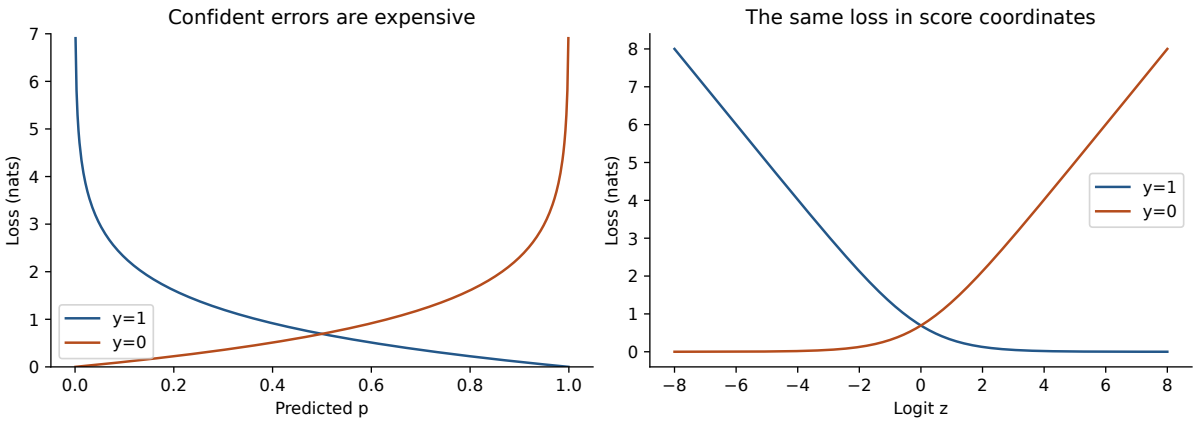


Figure 1.3: Cross-entropy in probability coordinates (left) and logit coordinates (right). The probability loss diverges at a wrong endpoint. In logit coordinates the wrong-direction loss grows approximately linearly for large magnitude.

The lecture's two-prediction question. For labels $(0, 1)$, prediction $(1, 1)$ assigns zero joint probability to the observed data. Its mean BCE is infinite, because the first observation has $-\log(1-1) = +\infty$. Prediction $(0.5, 0.5)$ assigns joint probability 0.25, and its mean loss is $-\frac{1}{2}\log(0.25) = \log 2 \approx 0.6931$. Cross-entropy therefore prefers the second prediction. Exact endpoints are not produced by a finite mathematical logit, but can occur through rounding. With $(1-\varepsilon, 1-\varepsilon)$ instead, the mean loss is $-\frac{1}{2}\log[\varepsilon(1-\varepsilon)]$, which diverges as $\varepsilon \downarrow 0$. At an endpoint, interpret zero-weight log terms by limits; do not compute $0 \log 0$ directly in floating-point code.

1.1.10 Why the best probability is honest, not always extreme

For a single observed positive label the loss prefers $p \to 1$. Does this mean training should always produce extreme probabilities? Consider repeated outcomes at the same input with true success probability q. The *expected* loss of reporting p is

$$r(p; q) = -q \log p - (1-q) \log(1-p).$$

For $0 < q < 1$,

$$\frac{\partial r}{\partial p} = -\frac{q}{p} + \frac{1-q}{1-p} = \frac{p-q}{p(1-p)}.$$

It is negative for $p < q$, zero for $p = q$, and positive for $p > q$. Hence the unique minimum is the true conditional probability $p = q$. For $q = 0$ or 1, the minimum is at the corresponding endpoint, approached as a limit by finite logits. This property is called a *strictly proper scoring rule*.

Suppose ten otherwise identical conditions yield seven successes and three failures. Their mean loss is $-0.7 \log p - 0.3 \log(1-p)$, minimized at $p = 0.7$, with logit $\log(7/3) \approx 0.8473$. Memorizing the latest outcome is not the same as modeling the underlying probability.

The irreducible minimum is $r(q; q) = H(q)$, and the excess expected loss is

$$r(p; q) - H(q) = q \log \frac{q}{p} + (1-q) \log \frac{1-q}{1-p} = D_{\mathrm{KL}}(\mathrm{Bern}(q) \,\|\, \mathrm{Bern}(p)) \geq 0.$$

The nonnegativity also follows from the minimization just proved. This gives a bridge to Section 1.2: cross-entropy is entropy plus the cost of using the wrong probabilities. A one-hot observed label is an empirical point mass, whose cross-entropy with the model is precisely the single-observation negative log-likelihood.

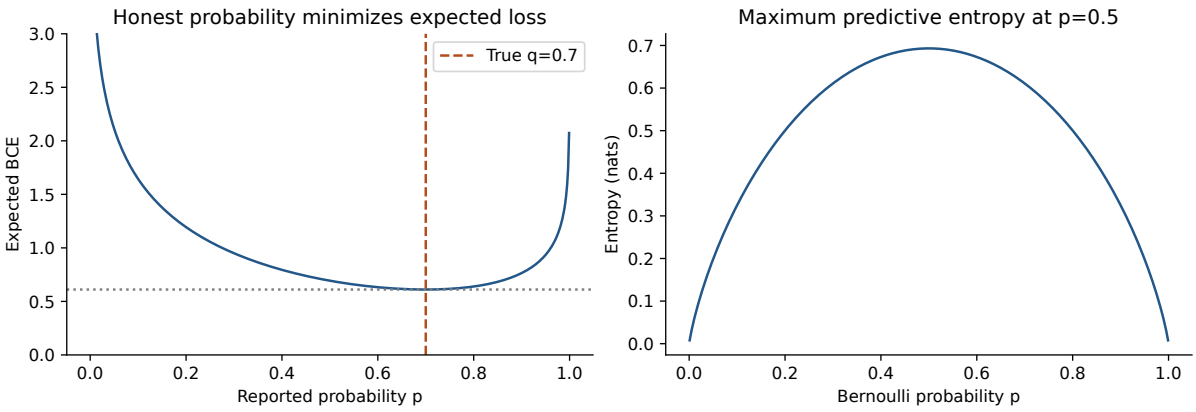


Figure 1.4: Left: when the true success frequency is 0.7, reporting 0.7 minimizes expected loss. The dotted horizontal line is the irreducible entropy. Right: entropy peaks at 0.5, quantifying uncertainty in a Bernoulli prediction.

Empirical versus population risk. Training minimizes a sample average. The quantity we ultimately care about is

$$R(w) = \mathbb{E}_{X,Y}[\ell(p_w(X), Y)] = \mathbb{E}_X[-q(X)\log p_w(X) - (1-q(X))\log(1-p_w(X))]\,.$$

This follows by first averaging over the two values of Y at each fixed X, then averaging over inputs. The weights are fixed while this expectation is evaluated; they vary when comparing models during optimization. Finite data, misspecified affine log-odds, and distribution shift can keep p_w far from q. A proper loss encourages honest probabilities in the population; it does not promise that every fitted finite-sample model is calibrated.

1.1.11 Deriving the gradient: why the residual is $p - y$

We want to adjust each weight in the direction that reduces the mean loss. For one observation, apply the chain rule through $w \to z \to p \to \ell$:

$$\begin{aligned}
\frac{\partial \ell}{\partial p} &= -\frac{y}{p} + \frac{1-y}{1-p} = \frac{p-y}{p(1-p)}, \\
\frac{\partial p}{\partial z} &= p(1-p), \\
\frac{\partial \ell}{\partial z} &= p - y, \\
\frac{\partial z}{\partial w_j} &= \tilde{x}_j, \qquad \boxed{\frac{\partial \ell}{\partial w_j} = (p-y)\tilde{x}_j.}
\end{aligned}$$

The apparent complicated fractions cancel because of the sigmoid–Bernoulli pairing. In particular, even when a confidently wrong sigmoid saturates, the BCE gradient with respect to its logit does not vanish: it approaches -1 for a very wrong positive label and $+1$ for a very wrong negative label.

For $y = 1$ and $p = 0.2$, $p - y = -0.8$. Gradient descent increases the score in the direction of this input, making the observed success more probable. More precisely, for a single-sample update $w^+ = w - \eta(p-y)\tilde{x}$,

$$z^+ = (w^+)^\top \tilde{x} = z + \eta(y-p)\|\tilde{x}\|^2.$$

For a positive label this increases its score; for a negative label it decreases it. Other examples may pull in conflicting directions, which is why learning requires a compromise rather than independent adjustments to every prediction.

Matrix form and dimensions. Let $X \in \mathbb{R}^{N\times(d+1)}$ have row n equal to $\tilde{x}_n^\top$, and let $y, p, z \in \mathbb{R}^N$. Then

$$z = Xw, \qquad p = \sigma(z) \quad \text{elementwise}, \qquad \boxed{\nabla J(w) = \frac{1}{N} X^\top (p - y).}$$

The gradient has shape $(d+1)$: $(d+1) \times N$ times $N \times 1$. The bias gradient is $N^{-1}\sum_n (p_n - y_n)$ because column zero of X is one. A full-batch update is

$$w_{t+1} = w_t - \eta \frac{1}{N} X^\top (\sigma(Xw_t) - y).$$

All predictions in this update are evaluated at the *current* weights w_t. Predictions change after the update. For a uniformly sampled minibatch of size B, replace the full average by its batch average. This is an unbiased estimator of the empirical gradient under uniform sampling. SGD is one possible solver, not part of the definition of logistic regression.

One complete update by hand. Take two observations $x^{(1)} = (0, 0), y_1 = 0$ and $x^{(2)} = (1, 2), y_2 = 1$, and initialize all three weights to zero. Both probabilities are 0.5, so

$$X = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 2 \end{bmatrix}, \quad p - y = \begin{bmatrix} 0.5 \\ -0.5 \end{bmatrix}, \quad \nabla J = \begin{bmatrix} 0 \\ -0.25 \\ -0.5 \end{bmatrix}.$$

With $\eta = 1$, $w^+ = (0, 0.25, 0.5)^\top$. The scores become $(0, 1.25)$ and the probabilities $(0.5, 0.7773)$. The mean loss decreases from 0.6931 to $\frac{1}{2}[0.6931 + \log(1 + e^{-1.25})] \approx 0.4725$. This separable toy dataset has no finite unregularized optimum; it illustrates an update, not convergence to finite weights.

1.1.12 Convexity: a proof, and what it does not promise

Differentiate the gradient again. Since $\partial p_n / \partial w = p_n(1 - p_n)\tilde{x}_n$,

$$\nabla^2 J(w) = \frac{1}{N} X^\top D X, \qquad D = \operatorname{diag}\big(p_n(1 - p_n)\big). \tag{1.6}$$

For any $v \in \mathbb{R}^{d+1}$,

$$v^\top \nabla^2 J(w) v = \frac{1}{N} \sum_{n=1}^{N} p_n(1 - p_n)(\tilde{x}_n^\top v)^2 \geq 0.$$

Thus the Hessian is positive semidefinite and J is convex in w. Every finite local minimum is global. This proof still works for any *fixed* nonlinear feature map, since its design matrix remains fixed. It does not establish convexity for a multilayer network whose feature representation is trained too.

For finite logits, all diagonal entries of D are positive. If X has full column rank, the Hessian is positive definite and J is strictly convex at every finite w, giving at most one finite minimizer. If columns are redundant, there may be whole flat directions. Strict convexity does not by itself guarantee that a minimum is attained.

The separability trap. Write $t_n = 2y_n - 1 \in \{-1, 1\}$. If some v strictly separates all training examples, $t_n \tilde{x}_n^\top v > 0$ for every n, then

$$J(cv) = \frac{1}{N} \sum_n \log(1 + e^{-c t_n \tilde{x}_n^\top v}) \longrightarrow 0 \quad (c \to \infty).$$

For any finite weights each term is strictly positive. The loss has infimum zero but no finite minimizer. Classification can become perfect while weight norms keep increasing. Quasi-separation creates related nonexistence issues. Therefore "convex" is not equivalent to "a unique finite solution exists."

A common remedy is an L^2 penalty on slopes,

$$J_\lambda(w) = J(w) + \frac{\lambda}{2}\|a\|_2^2, \qquad \nabla J_\lambda = \nabla J + \lambda(0, a^\top)^\top.$$

It discourages arbitrarily steep confidence transitions. With both classes present and fixed finite features, penalizing all slopes with $\lambda > 0$ also controls the unpenalized intercept and gives a finite unique optimum. If every label is one class, an unpenalized bias can still diverge. Penalizing all coordinates, including the bias, makes the objective coercive and strongly convex for $\lambda > 0$, guaranteeing a unique finite minimizer. The penalty changes the estimation objective: it is no longer unregularized maximum likelihood. Its strength must be interpreted relative to sum versus mean loss.

Choosing a learning rate from curvature. Because $p_n(1 - p_n) \le 1/4$,

$$0 \preceq \nabla^2 J(w) \preceq \frac{1}{4N} X^\top X, \qquad L = \frac{\|X\|_2^2}{4N}$$

is a global gradient-Lipschitz bound (spectral norm). Smoothness gives

$$J(w - \eta g) \le J(w) - \eta(1 - L\eta/2)\|g\|^2, \qquad g = \nabla J(w).$$

Thus $0 < \eta < 2/L$ yields descent when $g \neq 0$; $\eta \le 1/L$ is a convenient conservative choice. Add λ to the bound for the stated L^2 penalty. This full-batch bound does not mean every stochastic step decreases the loss. Standardizing feature scales often improves conditioning and makes the optimization more efficient.

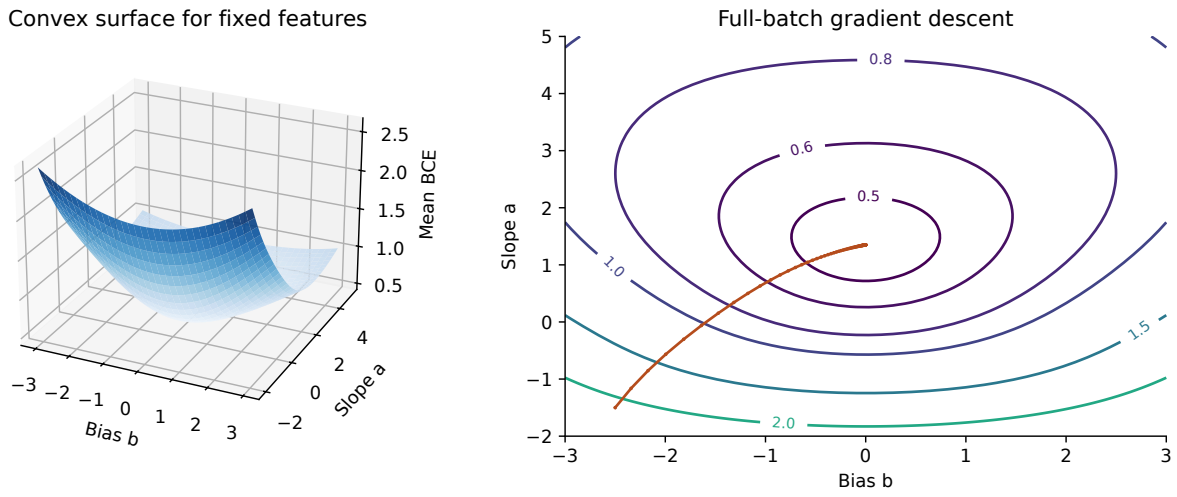


Figure 1.5: Mean BCE for a nonseparable one-feature dataset, with bias and slope as the two axes. The contour plot shows gradient descent from $(-2.5, -1.5)$. The surface and the contours describe the same objective in parameter space, not the decision boundary in input space.

1.1.13 Computational complexity: what exactly is being computed?

For dense data with N observations and d raw features:

Operation	Output	Time
Augmented matrix–vector product Xw	N logits	$\Theta(N(d+1))$
Elementwise sigmoid	N probabilities	$\Theta(N)$
Loss terms and reduction, given logits	One scalar	$\Theta(N)$
Gradient product $X^\top(p-y)$	$d+1$ derivatives	$\Theta(N(d+1))$

So computing BCE *from raw inputs and weights* is $\Theta(Nd)$ for $d \geq 1$. If probabilities have already been computed and the question only counts the loss reduction, it is $\Theta(N)$. A full-batch gradient step is also $\Theta(Nd)$; T steps cost $\Theta(TNd)$. We treat scalar arithmetic and transcendental functions as constant-cost operations in this usual model.

Storing dense X costs $\Theta(Nd)$ memory, and vectorized predictions need $\Theta(N)$ additional memory. Minibatches can reduce the working prediction storage to $\Theta(B)$; streaming avoids storing the full dataset. Sparse matrix products can cost $\Theta(\mathrm{nnz}(X)+N)$ instead. Explicitly forming a dense Hessian costs $O(Nd^2)$ time and $O(d^2)$ extra space, which explains why first-order methods are attractive when d is large.

1.1.14 Numerical stability: evaluate the mathematical quantity safely

The direct expression $-y\log\sigma(z) - (1-y)\log(1-\sigma(z))$ is fragile in floating point. A large positive score can round $\sigma(z)$ to one, and then $\log(1-p)$ becomes $\log 0$. Large negative scores can overflow e^{-z}. These are numerical failures, not infinite mathematical losses at finite z.

Use identities before computing. Since

$$\log\sigma(z) = z - \log(1+e^z), \qquad \log(1-\sigma(z)) = -\log(1+e^z),$$

substitution into BCE yields

$$\ell(z,y) = \log(1+e^z) - yz = \mathrm{softplus}(z) - yz. \tag{1.7}$$

A stable softplus is $\max(z,0) + \log(1+e^{-|z|})$. For binary labels an even cleaner form is

$$\boxed{\ell(z,y) = \log(1+e^{(1-2y)z}) = \mathrm{softplus}(-tz), \quad t = 2y-1.}$$

The scalar tz is the signed margin: a large positive margin means confidently correct, a negative margin means wrong. NumPy's `logaddexp(0, u)` computes $\log(1+e^u)$ stably. The signed-margin form also avoids subtracting two large nearly equal numbers for a correctly classified positive example.

```python
import numpy as np

def sigmoid(z):
    z = np.asarray(z, dtype=float)
    return np.exp(-np.logaddexp(0.0, -z))

def loss_and_gradient(X, y, w):
    # X: (N, d+1), y: (N,), w: (d+1,)
    z = X @ w
    loss = np.logaddexp(0.0, (1.0 - 2.0*y)*z).mean()
    gradient = X.T @ (sigmoid(z) - y) / len(y)
    return loss, gradient
```

At $z = -1000, y = 1$, the loss is approximately 1000, not a NaN. Clipping probabilities to $[\varepsilon, 1-\varepsilon]$ can prevent a log failure, but changes the objective near endpoints and can make a stated gradient inconsistent with that clipped objective. Computing from logits is preferable when available. Keep y one-dimensional with the code above: mixing shape $(N,1)$ labels and $(N,)$ probabilities can accidentally broadcast to an (N,N) array.

1.1.15 Worked robotics setup: grasp success, cost, and evaluation

We now connect the equations to an entire modeling workflow. Suppose a robot uses two nonnegative measurements: centering error e in millimeters and angular error θ in degrees. For an interpretable synthetic example, assume the true conditional success probability is

$$q(e, \theta) = \sigma(3 - 0.25e - 0.08\theta). \tag{1.8}$$

These coefficients are illustrative, not measured hardware parameters. Real success also depends on object geometry, friction, contact, and perception. The model says that one extra millimeter multiplies success odds by $e^{-0.25} \approx 0.779$, while one extra degree multiplies them by $e^{-0.08} \approx 0.923$, holding the other feature fixed.

At $e = 4, \theta = 10$, the logit is 1.2 and $q \approx 0.7685$. At $e = 8, \theta = 20$, it is -0.6 and $q \approx 0.3543$. With equal error costs, the success boundary is

$$0.25e + 0.08\theta = 3.$$

This line is a contour of equal probability, not a physical law of grasping.

Derive the action threshold from consequences. Suppose accepting a grasp that fails costs C_{FP}, while rejecting a grasp that would succeed costs C_{FN}. Assume the two correct decisions have zero cost. Given a success probability p, the expected costs are

$$\text{accept: } C_{\mathrm{FP}}(1 - p), \qquad \text{reject: } C_{\mathrm{FN}}p.$$

Accept if the first is smaller, giving

$$p \geq \tau = \frac{C_{\mathrm{FP}}}{C_{\mathrm{FP}} + C_{\mathrm{FN}}}, \qquad z \geq \log \frac{\tau}{1 - \tau}. \tag{1.9}$$

With $C_{\mathrm{FP}} = 8$ and $C_{\mathrm{FN}} = 1$, $\tau = 8/9 \approx 0.8889$ and the score threshold is $\log 8 \approx 2.0794$. Equation (1.8) then permits acceptance only when

$$0.25e + 0.08\theta \leq 3 - \log 8 \approx 0.9206.$$

The earlier 0.7685 example would pass a 0.5 rule but fail this cost-sensitive rule. The probability model has not changed; the action criterion has. In deployment this rule relies on useful probability estimates and appropriate costs. It is not a safety guarantee.

A complete reproducible experiment. The companion script generates 1,200 trials using seed 7, with independent $e \sim U(0, 20)$, $\theta \sim U(0, 40)$ and $Y \mid e, \theta \sim \mathrm{Bern}(q(e, \theta))$. It randomly splits them into 720 training, 240 validation, and 240 test observations. This random split is appropriate for this independently generated simulation. On real robot logs, split by object, session, or time as appropriate so near-duplicate frames and trials do not leak across sets.

Compute each feature's mean μ_j and standard deviation s_j on the training set only. Transform all sets using $u_j = (x_j - \mu_j)/s_j$ and prepend the bias column. Fit the standardized logistic model using full-batch gradient descent and a fixed illustrative slope penalty $\lambda = 0.02$. The script uses the curvature bound above to choose a conservative step size, with 1,800 updates. A practical study would select the penalty using validation data too.

Standardization changes coordinates, not the expressive family. If the learned score is $b_s + \sum_j a_{s,j}(x_j - \mu_j)/s_j$, its physical-unit coefficients are

$$a_j = \frac{a_{s,j}}{s_j}, \qquad b = b_s - \sum_j \frac{a_{s,j}\mu_j}{s_j}.$$

However, an L^2 penalty in standardized coordinates does change the implicit penalty on physical-unit coefficients. This is one reason scaling is a modeling choice as well as a numerical convenience.

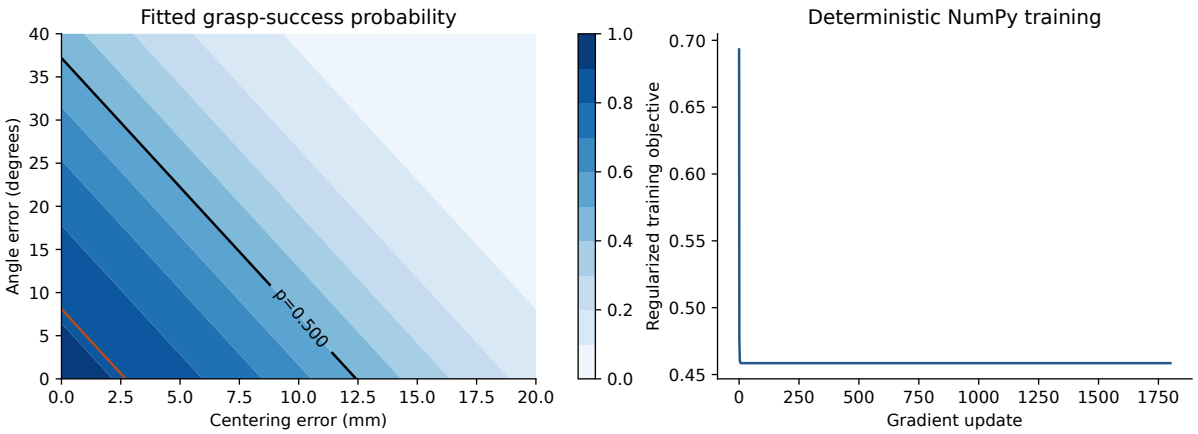


Figure 1.6: Synthetic robot experiment. Left: the fitted success field in physical units, with the 0.5 and 8/9 thresholds. Right: the regularized training objective decreases under the chosen full-batch step size.

Evaluate probabilities and decisions separately. Report unpenalized test BCE to measure probability quality, accuracy for a stated threshold, and empirical decision cost for the actual action criterion. A reliability diagram groups predicted probabilities into bins and compares each bin's mean probability with its observed success frequency. Small bins are noisy; an apparent deviation from the diagonal is not automatically systematic miscalibration. The script records each bin's sample count.

The script also selects a threshold from $0.05, 0.06, \ldots, 0.95$ using validation cost alone, then evaluates it on the test data. An empirically selected threshold can differ from 8/9 because of finite samples and imperfect probabilities. Do not select it by looking at test results: the test set is for the final assessment. For this seeded run the unpenalized test BCE is 0.4317 nats and accuracy at 0.5 is 80.0%. The validation-selected threshold is 0.84.

Decision threshold	Mean test cost
0.5	0.7542
8/9 (cost-derived)	0.3250
0.84 (validation-selected)	0.3708

Here validation tuning does not beat the cost-derived threshold on the test sample. This is a useful reminder that selecting a threshold on a finite validation set is itself an estimation problem. The full numerical output is in the adjacent `results.json`; all reported quantities can be regenerated from the code.

What would break this setup? A new gripper or different object material can change $q(x)$ even when the two features look familiar. Missing interactions may make affine log-odds inadequate. Selection bias arises if training only records grasps the old policy chose to attempt; rejected grasps may have no observed outcome. Label errors, rare failures, correlated trials, and feature drift also matter. More confident numbers alone solve none of these problems. The model's meaning is conditional on its observations, data collection, and deployment distribution.

1.1.16 From-scratch practice and checks

The executable companion is `deep_learning/code/logreg_1_1.py`. It uses NumPy for the model and optimization, and Matplotlib for figures; it uses no machine-learning library. Run it from the repository root:

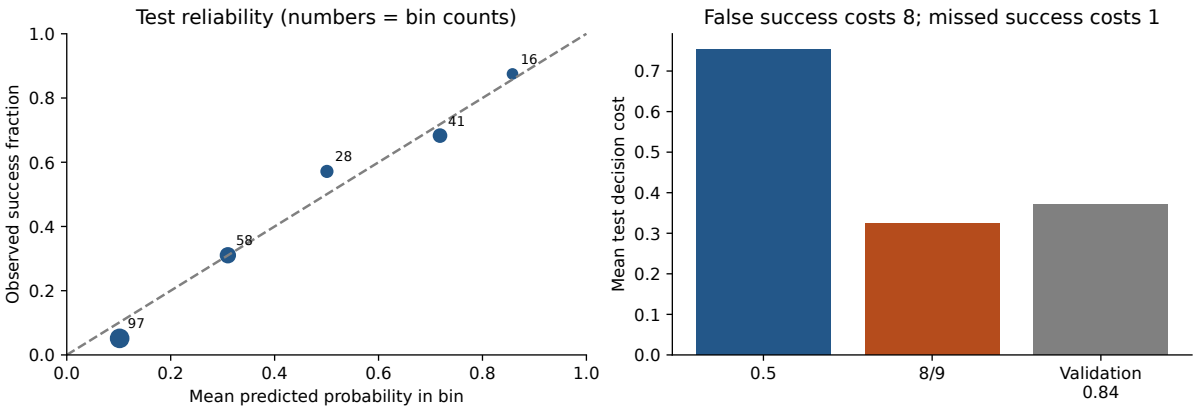


Figure 1.7: Held-out assessment for synthetic grasp trials. Left: a reliability diagram, with counts showing how much evidence supports each bin. Right: mean cost at a default threshold, the analytically derived cost threshold, and a validation-selected threshold. These are one simulation's measurements, not claims of general superiority or hardware performance.

```
python deep_learning/code/logreg_1_1.py
make dl
```

It regenerates seven vector-PDF figures and a JSON result summary in `deep_learning/figures/logreg_1_1/`. The chapter can be compiled with the committed figures without running Python. The full script includes an L^2-regularized objective, training, data splitting, standardization, evaluation, and the following meaningful numerical checks:

1. Compare the analytic gradient against centered finite differences, $[J(w + \varepsilon e_j) - J(w - \varepsilon e_j)]/(2\varepsilon)$.
2. Check the Hessian's eigenvalues on a small dataset and verify that the chosen full-batch updates do not increase the regularized objective.
3. Evaluate extreme logits ± 1000 and require finite losses and probabilities.
4. Check $\sigma(0) = 1/2$ and $\sigma(-z) = 1 - \sigma(z)$.

These checks catch implementation mistakes; they do not replace the mathematical proofs or establish generalization to a robot.

Suggested implementation order. First reproduce the two-point hand update using only arrays. Then implement stable sigmoid and BCE, verify the gradient numerically, add repeated updates, and plot the training loss. Only after that add standardization, a validation split, and decision costs. Predict what each change should do before running it. This makes the code an experiment on the mathematical ideas rather than a recipe.

1.1.17 Reasoning exercises

Try these before reading the worked answers below. They are independent of the main derivation, so the conceptual narrative need not be interrupted.

Exercise 1.1 (Same classes, different probabilities)**.** For $w = (-3, 1, 1)^\top$, compare w and $10w$ at $(1, 2)$ and $(2, 2)$. Do the boundaries, hard predictions, and losses for a positive label agree?

Exercise 1.2 (Recover a probability model)**.** An intercept-only dataset contains 20 successes and 30 failures. Derive the maximum-likelihood probability and intercept. What happens if all 50 succeed?

Exercise 1.3 (Costs change a boundary)**.** A classifier has score $z = -3 + x_1 + x_2$. Derive the action boundary when a false positive costs nine times a false negative. Which side predicts one?

Exercise 1.4 (Feature maps and convexity)**.** Use features (x_1, x_2, x_1x_2) to express XOR on $\{-1, 1\}^2$. Explain why the training objective is convex in its coefficients, despite the nonlinear boundary. Would the same proof apply if the features were outputs of a trained hidden layer?

Exercise 1.5 (Complexity and data shape)**.** For $N = 10^6$ and $d = 100$, identify the dominant forward-pass operation. Explain why storing labels as $(N, 1)$ while predictions have shape $(N,)$ is dangerous. How does a batch of size B change working memory and per-step time?

1.1.18 Worked answers and conceptual checkpoints

1. Scaling. At $(1, 2)$, both logits are zero and both probabilities are 0.5. At $(2, 2)$, the logits are 1 and 10, with probabilities about 0.7311 and 0.99995. Positive-label losses are $\log(1 + e^{-1}) \approx 0.3133$ and $\log(1 + e^{-10}) \approx 0.0000454$. Boundaries and hard classes agree. If the label at $(2, 2)$ were zero instead, the larger scale would be far more costly: about 10.00005 versus 1.3133. Confidence has consequences.

2. Intercept-only estimation. The loss is $-0.4 \log p - 0.6 \log(1 - p)$; differentiating gives the minimum $p = 0.4$. Thus $b = \log(0.4/0.6) = \log(2/3) \approx -0.4055$. If all labels are one, the loss decreases as $b \to +\infty$, and no finite unregularized optimum exists.

3. Costs. The threshold is $\tau = 9/10$, so the logit threshold is $\log 9$. The new boundary is $x_1 + x_2 = 3 + \log 9 \approx 5.1972$, with class one on the greater-than side. It is parallel to the equal-cost boundary and requires stronger evidence.

4. Fixed features. Take zero bias and zero coefficients on x_1, x_2, with coefficient $-k$ on x_1x_2, $k > 0$. The four signed scores have the required signs. The Hessian proof applies to the resulting fixed design matrix. With a trainable hidden layer, the feature matrix itself depends on parameters, and the positive semidefinite form above is no longer the full Hessian.

5. Computation. The dense product Xw uses roughly $N(d + 1)$ multiply–accumulate operations, about 10^8 here. Broadcasting $(N,)$ minus $(N, 1)$ can create an (N, N) array, which is both mathematically wrong and enormous. A dense batch step costs $\Theta(Bd)$ time and stores $\Theta(Bd)$ feature entries if batches are streamed, plus parameters and $\Theta(B)$ temporary predictions. An epoch still processes N observations and costs $\Theta(Nd)$.

Reconstruct the method from its meaning

An affine score describes additive log-odds. The sigmoid converts that evidence into a Bernoulli probability. Independence turns observed-label probabilities into a product likelihood, and the negative logarithm turns that product into an additive loss. The gradient $X^\top(p - y)/N$ adjusts evidence according to probability errors. Convexity follows from nonnegative Bernoulli variances, but finite solutions require more than convexity. Finally, the decision threshold comes from the consequences of actions, not from the probability model alone.

Scope and reading. This section expands the supplied lecture excerpts (slides 5–10 and their questions). Section 1.2 can develop entropy, KL divergence, and maximum likelihood more generally; the later sections can reinterpret the same affine–sigmoid computation as a neural-network unit. General background texts already present in this repository include Bishop [1] and Goodfellow et al. [2]. All plots in this section are generated by its companion script from explicit formulas or synthetic data.

1.2 Cross Entropy, KL Divergence, and Maximum Likelihood

Three quantities and three questions

Entropy asks: how unpredictable is the source itself? Cross-entropy asks: how surprised will a particular model be by that source? KL divergence asks: how much extra surprise does using that model create? Maximum likelihood chooses the model that assigns the greatest joint probability, or density, to the observed data. These ideas connect through one operation: averaging negative log probabilities under the distribution that supplies the observations.

1.2.1 Start with a robot that must describe what it sees

A mobile robot sends a stream of corridor events: clear path, person, cart, or blocked path. Suppose their long-run probabilities are

$$P = (1/2, 1/4, 1/8, 1/8).$$

Sending two bits for every event works: there are four possibilities. But the outcomes are not equally common. Could we give frequent outcomes shorter messages and rare ones longer messages, reducing the average communication cost?

Now imagine the robot's model predicts a different frequency distribution Q. Even if the communication channel is perfect, a code designed for Q can be wasteful on events actually drawn from P. This is a concrete way to separate *unavoidable uncertainty* from *avoidable model mismatch*. The same distinction applies to prediction: a noisy task can have a positive best possible log loss, while a confident but wrong model adds further loss.

The narrative in this section is

$$\text{surprise} \longrightarrow \text{average surprise} \longrightarrow \text{mismatch} \longrightarrow \text{estimating a model from data.}$$

We first work on a finite outcome space. Continuous measurements need additional care and are treated explicitly later. Throughout, P is the distribution that supplies outcomes and Q is the distribution used to score them, unless an expression explicitly swaps their roles.

1.2.2 Why should information be a negative logarithm?

Suppose an event has probability p. We seek a numerical measure $I(p)$ of the information gained when learning that it occurred. A plausible set of requirements is:

1. A certain event gives no new information: $I(1) = 0$.
2. A rarer event is more informative: $I(p)$ decreases as p increases.
3. For independent events, information adds: $I(pq) = I(p) + I(q)$.
4. The measure varies continuously with probability.

The independence clause is essential: probabilities multiply for independent events. For dependent events, the relevant factorization is $P(A, B) = P(A)P(B \mid A)$, so information adds using a conditional probability.

To see what these requirements imply, set $p = e^{-t}$ and define $F(t) = I(e^{-t})$ for $t \geq 0$. Then

$$F(t + s) = F(t) + F(s).$$

For positive integers, repeated addition gives $F(n) = nF(1)$. Dividing an interval into equal parts gives $F(m/n) = (m/n)F(1)$ for nonnegative rationals. Continuity extends this to real $t \geq 0$. Therefore $F(t) = ct$, with $c > 0$ by the rarity requirement, and

$$\boxed{I(p) = -c \ln p.}$$

The choice of c fixes the unit. With base-two logarithms, a fair coin outcome has information $-\log_2(1/2) = 1$ bit. With natural logarithms the unit is a nat. Since $\log_2 p = \ln p / \ln 2$, a value in nats divided by $\ln 2$ is its value in bits. Information here is about reduction of probabilistic uncertainty, not the semantic importance of a message. A rare sensor glitch can have high surprise without being useful for navigation.

One event versus a distribution. For $P = (1/2, 1/4, 1/8, 1/8)$, the four event surprises are $1, 2, 3, 3$ bits. A particular observed event has one of these values. Entropy will be the average of those values before we know which event will occur. Do not confuse the information of an outcome with the uncertainty of the whole distribution.

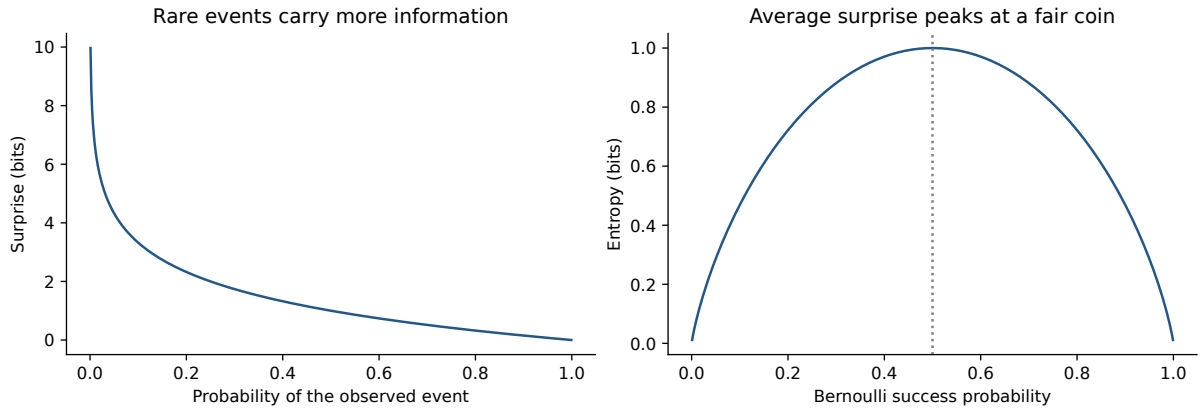


Figure 1.8: Left: surprise of one event grows as its probability decreases. Right: the average surprise of a Bernoulli source is largest at equal probabilities. These curves answer different questions: event surprise and source uncertainty.

1.2.3 Expectation turns surprise into entropy

For a finite random variable X with outcomes $x_1, \ldots, x_K$, let $p_k = P(X = x_k)$. By the definition of expectation, any function $f(X)$ has mean $\sum_k p_k f(x_k)$. Choosing $f(x) = -\log P(x)$ gives

$$\boxed{H(P) = \mathbb{E}_{X \sim P}[-\log P(X)] = -\sum_{k=1}^{K} p_k \log p_k.} \tag{1.10}$$

The weights outside the logarithm describe *how often* an event occurs; the logarithm describes *how surprising* that event is. Entropy is not $-\log \mathbb{E}[P(X)]$, and not an unweighted average over the list of possible outcomes. Moving a nonlinear logarithm outside an expectation changes the quantity.

For the robot stream, in bits,

$$H(P) = \tfrac{1}{2}(1) + \tfrac{1}{4}(2) + \tfrac{1}{8}(3) + \tfrac{1}{8}(3) = 1.75.$$

This is below the two bits needed by an equiprobable four-outcome source. A deterministic source has entropy zero: it never surprises an observer who knows the distribution. On a K-outcome space,

$$0 \le H(P) \le \log K,$$

with the upper bound attained by the uniform distribution. We will prove the upper bound from KL nonnegativity below. Zero-probability events contribute zero by the continuous extension $\lim_{t\downarrow 0} t \log t = 0$.

A thought experiment. Imagine predicting the next symbol from a fair coin source versus a coin that comes up heads 99% of the time. For the biased coin, usually there is almost nothing new to learn, but occasionally a tail is very surprising. Entropy balances these two facts through a frequency-weighted average. It does not simply count the number of logically possible outcomes.

1.2.4 Coding makes the average operational

A binary *prefix code* is a collection of bit strings in which no complete codeword is the prefix of another. This allows concatenated messages to be decoded without separators. For the robot stream, consider these three codes:

Event	True P	P-matched code	Uniform code	R-matched code
Clear	1/2	0	00	110
Person	1/4	10	01	111
Cart	1/8	110	10	10
Blocked	1/8	111	11	0

The last code matches $R = (1/8, 1/8, 1/4, 1/2)$. All three are valid prefix codes; they simply allocate short words differently. Their expected lengths under P are, respectively,

$$1.75, \qquad 2, \qquad \tfrac{1}{2}(3) + \tfrac{1}{4}(3) + \tfrac{1}{8}(2) + \tfrac{1}{8}(1) = 2.625 \text{ bits}.$$

The first code is most efficient for this source. The last makes the common "clear" event expensive and saves bits on a rare event instead.

Why ideal lengths are negative log probabilities. A complete binary code with lengths ℓ_k satisfies $\sum_k 2^{-\ell_k} = 1$. Think of each prefix as selecting a fraction $2^{-\ell_k}$ of the infinite binary tree. Assigning probability q_k the fraction q_k suggests the ideal length $\ell_k = -\log_2 q_k$. The code tree has only integer depths, however; these ideal lengths need not be integers. The dyadic probabilities in the table were chosen so they are.

More generally, the Kraft inequality says prefix-code lengths satisfy $S = \sum_k 2^{-\ell_k} \le 1$. Set $r_k = 2^{-\ell_k}/S$. Then, in bits,

$$\mathbb{E}_P[\ell_X] = -\sum_k p_k \log_2 r_k - \log_2 S = H(P) + D_{\mathrm{KL}}(P\|R) - \log_2 S \ge H(P).$$

This lower bound uses the KL nonnegativity proved shortly. Conversely, choosing $\ell_k = \lceil -\log_2 p_k \rceil$ gives a prefix code with

$$H(P) \le \mathbb{E}_P[\ell_X] < H(P) + 1$$

when the source has positive probabilities on its supported outcomes. For blocks of n independent events, applying the construction to the block probabilities gives an overhead of less than one bit *per block*, hence less than $1/n$ bit per event.

Read the slide's coding statement precisely

Entropy is an ideal average code length and an asymptotically achievable rate for an i.i.d. source, not necessarily the exact length of a one-symbol binary code. A Bernoulli source can have entropy 0.2 bits even though a single nontrivial binary symbol needs an integer-length codeword. Block coding is how fractional rates become meaningful. Dependencies require conditional or joint coding; the marginal entropy alone need not be the best rate for a sequence.

1.2.5 Cross-entropy: keep the source, change the scoring model

Suppose outcomes still come from P, but we evaluate them using probabilities $q_k = Q(X = x_k)$. The average model-assigned surprise is

$$\boxed{H(P, Q) = \mathbb{E}_{X\sim P}[-\log Q(X)] = -\sum_k p_k \log q_k.} \tag{1.11}$$

The first argument determines the averaging weights. The second determines the logarithmic score. Swapping the arguments changes both roles. This distinction is the single most useful way to read a cross-entropy expression.

For the robot source and a uniform model, every event receives two bits of surprise, so $H(P, Q) = 2$. For R above, $H(P, R) = 2.625$. In both cases the actual source remains P and has the same entropy 1.75.

For a general model Q with positive probability wherever P has positive probability, the ideal Q-based length is $-\log_2 q_k$. Rounded Shannon lengths have expected length between $H(P, Q)$ and $H(P, Q) + 1$ bits. This is an ideal or asymptotic mismatched-coding interpretation; an arbitrary integer code described as "optimized for Q" need not have length exactly $-\log_2 q_k$.

Support is part of the meaning. If $p_k > 0$ but $q_k = 0$, an event that really can happen has been declared impossible. Its log loss is infinite, so $H(P, Q) = +\infty$. If $p_k = 0$, that outcome is never scored under P and its term is defined to contribute zero. Do not evaluate $0 \log 0$ naively in numerical code.

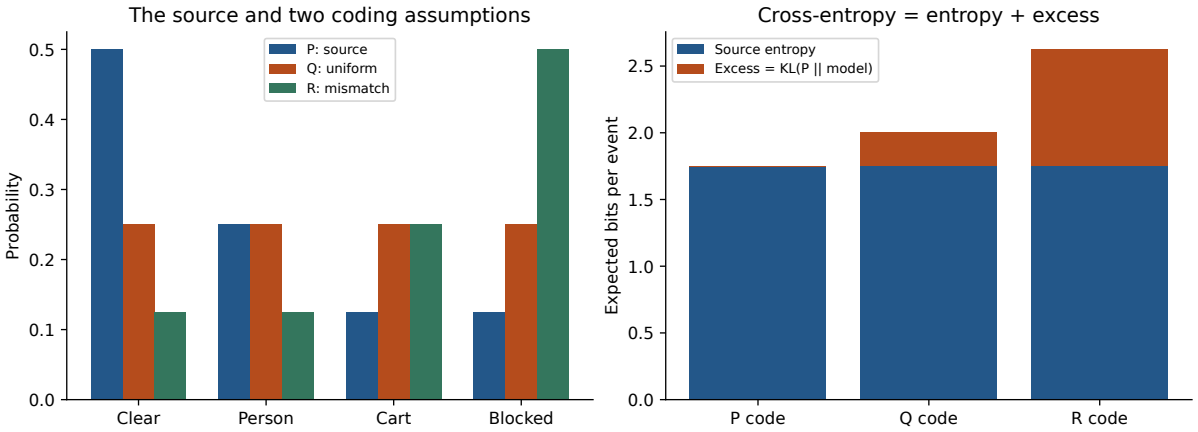


Figure 1.9: The same source is scored using three models. The unavoidable source entropy is unchanged. The additional orange part of each expected code length is KL divergence. The probabilities are dyadic, so the displayed coding costs are exact for the prefix codes in the table.

1.2.6 KL divergence is the excess expected log loss

Subtract the source entropy from cross-entropy, assuming the expressions are finite, and collect the logarithms:

$$\begin{aligned} H(P,Q) - H(P) &= \sum_k p_k(\log p_k - \log q_k) \\ &= \sum_k p_k \log \frac{p_k}{q_k} = \mathbb{E}_{X\sim P}\left[\log \frac{P(X)}{Q(X)}\right] =: D_{\mathrm{KL}}(P\|Q). \end{aligned} \tag{1.12}$$

Equivalently, $D_{\mathrm{KL}}(P\|Q) = -\mathbb{E}_P \log[Q(X)/P(X)]$, which matches the slide. Thus

$$\boxed{H(P,Q) = H(P) + D_{\mathrm{KL}}(P\|Q).}$$

For infinite quantities, do not treat $+\infty - \infty$ as a legitimate subtraction; define KL through the log-ratio expression with suitable support and integrability conditions.

Each individual log ratio may be positive or negative: a model can assign *more* probability than P to an outcome. KL is the average over P. Normalization forces a tradeoff: giving extra probability to some outcomes leaves less for others. The average cannot be negative.

Proof of nonnegativity, including zero probabilities. The elementary inequality $\log t \le t - 1$ for $t > 0$ follows by minimizing $t - 1 - \log t$: its derivative is $1 - 1/t$, and its minimum at $t = 1$ is zero. Let $S = \{k : p_k > 0\}$, and first suppose $q_k > 0$ on S. Applying the inequality to $t = q_k/p_k$ gives

$$\begin{aligned} -D_{\mathrm{KL}}(P\|Q) &= \sum_{k\in S} p_k \log \frac{q_k}{p_k} \\ &\le \sum_{k\in S} p_k \left(\frac{q_k}{p_k} - 1\right) = \sum_{k\in S} q_k - 1 \le 0. \end{aligned}$$

Equality requires $q_k = p_k$ for every $k \in S$ and no probability mass outside S; hence $P = Q$. If Q has a zero on positive P mass, KL is infinite and the inequality still holds. This proves both nonnegativity and the equality case.

Consequences. The best possible predictive distribution under expected log loss is $Q = P$. Moreover, using the uniform distribution $U_k = 1/K$,

$$D_{\mathrm{KL}}(P\|U) = \sum_k p_k \log p_k + \log K = \log K - H(P) \ge 0,$$

proving the maximum-entropy bound. An unrestricted distribution family can attain zero KL; a restricted model family may not contain P and may have a positive minimum mismatch.

1.2.7 Why KL is directed and is not a distance metric

For a concrete example, use natural logs and $P = (0.8, 0.2)$, $Q = (0.5, 0.5)$. Then

$$\begin{aligned} D_{\mathrm{KL}}(P\|Q) &= 0.8 \ln 1.6 + 0.2 \ln 0.4 \approx 0.192745, \\ D_{\mathrm{KL}}(Q\|P) &= 0.5 \ln 0.625 + 0.5 \ln 2.5 \approx 0.223144. \end{aligned}$$

Different sampling distributions produce different averages. KL is not symmetric and therefore cannot be a metric; it also does not generally satisfy the triangle inequality. It is nonnegative and identifies equality, but those two properties alone do not make a geometric distance.

For an explicit triangle-inequality counterexample, take $A = (1, 0)$, $B = (1/2, 1/2)$, and $C = (1/4, 3/4)$. Then $D(A\|C) = \ln 4 \approx 1.3863$, while $D(A\|B) + D(B\|C) = \ln 2 + \frac{1}{2}\ln(4/3) \approx 0.8370$. The direct divergence exceeds the sum through the intermediate distribution.

The support asymmetry is even sharper. For $P = (1, 0)$ and $Q = (1/2, 1/2)$, $D_{\mathrm{KL}}(P\|Q) = \ln 2$, while $D_{\mathrm{KL}}(Q\|P) = +\infty$. Under P only the first event is sampled. Under Q the second event occurs with positive probability but is impossible under P.

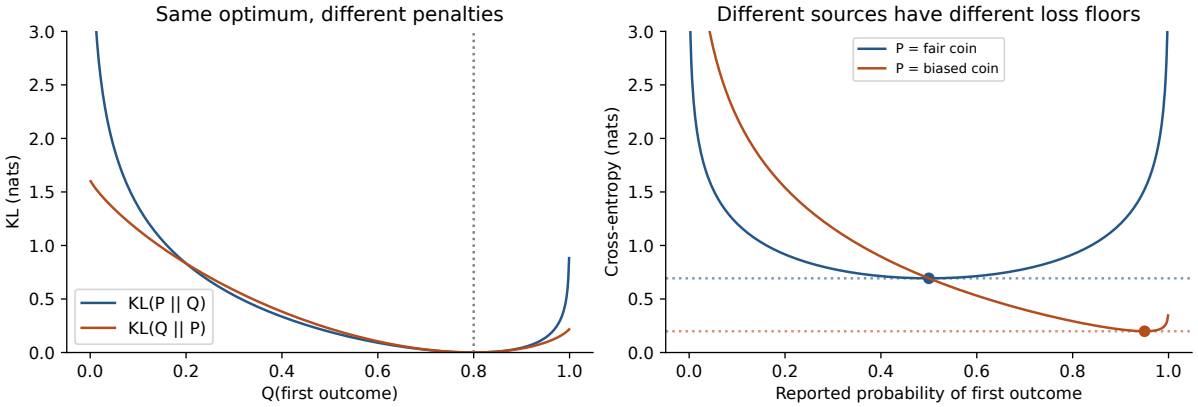


Figure 1.10: Left: the two directions have the same zero when the distributions match, but different penalties elsewhere. Right: different true sources have different minimum cross-entropies. Subtracting each source's entropy isolates model mismatch. Natural logs are used in both panels.

1.2.8 Conditioning: knowing the context changes uncertainty

A robot's observations are not drawn from one label distribution independently of its inputs. Let X denote the observed context and Y a semantic label. The relevant uncertainty is the conditional entropy

$$H_P(Y \mid X) = \mathbb{E}_{X \sim P_X}\left[-\sum_y P(y \mid X) \log P(y \mid X)\right].$$

We average uncertainty *within* each context over how often contexts occur. Using $P(x, y) = P(x)P(y \mid x)$ and taking negative logs and expectations gives the chain rule

$$H_P(X, Y) = H_P(X) + H_P(Y \mid X).$$

The expected information gained about Y from knowing X is

$$I_P(X; Y) = H_P(Y) - H_P(Y \mid X) = D_{\mathrm{KL}}(P_{XY}\|P_X P_Y) \geq 0.$$

This follows by expanding the joint log ratio and collecting the marginal and conditional terms. Conditioning reduces uncertainty *on average*; a particular unusual context can have more label entropy than the marginal.

For a conditional model $Q_\theta(y \mid x)$, expected log loss decomposes as

$$\underbrace{\mathbb{E}_{P_{XY}}[-\log Q_\theta(Y \mid X)]}_{\text{conditional cross-entropy}} = H_P(Y \mid X) + \mathbb{E}_{P_X} D_{\mathrm{KL}}(P(\cdot \mid X)\|Q_\theta(\cdot \mid X)). \tag{1.13}$$

To derive it, at each fixed x add and subtract $-\sum_y P(y \mid x) \log P(y \mid x)$, then average over x. This is the version most directly relevant to supervised learning.

1.2.9 Maximum likelihood: choosing a model after seeing data

We now move from two known distributions to an unknown source. Let $\mathcal{D} = (x^{(1)}, \ldots, x^{(m)})$ be observations drawn independently from a common but unknown distribution P_{data}. A parametric family Q_θ is a set of candidate distributions indexed by θ.

Before observing data, $Q_\theta(x)$ is a probability as a function of x for a fixed parameter. After observing the data, the same formula, considered as a function of θ, is a *likelihood*. The data are held fixed while we compare models:

$$\mathcal{L}(\theta; \mathcal{D}) = Q_\theta(\mathcal{D}), \qquad \hat{\theta}_{\text{ML}} \in \arg\max_\theta \mathcal{L}(\theta; \mathcal{D}).$$

We use membership in the argmax set because a maximizer might not be unique. There can also be no finite maximizer, as with separable unregularized logistic regression in Section 1.1. An *estimator* is a rule mapping a dataset to a parameter value; before sampling, it is itself random. Its realized output is an estimate.

Likelihood is not a posterior probability of θ. A posterior requires a prior through Bayes' rule. A high likelihood also does not by itself establish that the family is correct, or that the estimate will generalize.

Why independence appears in the derivation. Under the i.i.d. assumption,

$$Q_\theta(\mathcal{D}) = \prod_{i=1}^{m} Q_\theta(x^{(i)}).$$

For correlated observations, one instead needs an appropriate joint model, for example

$$Q_\theta(x^{(1)}) \prod_{i=2}^{m} Q_\theta(x^{(i)} \mid x^{(1:i-1)}).$$

The logarithm still turns a product into a sum, but the factors are conditional. Treating adjacent robot-video frames as independent is a modeling assumption, not an automatic consequence of having stored them in separate rows.

1.2.10 The empirical distribution: turning a dataset into an expectation

For discrete outcomes, define

$$\hat{P}_m(x) = \frac{1}{m} \sum_{i=1}^{m} \mathbf{1}\{x^{(i)} = x\}.$$

It is the observed relative frequency of each outcome. Repeated observations count repeatedly. For any function f,

$$\begin{aligned} \mathbb{E}_{\hat{P}_m}[f(X)] &= \sum_x \left[\frac{1}{m} \sum_i \mathbf{1}\{x^{(i)} = x\} \right] f(x) \\ &= \frac{1}{m} \sum_i \sum_x \mathbf{1}\{x^{(i)} = x\} f(x) = \frac{1}{m} \sum_i f(x^{(i)}). \end{aligned}$$

This is why a sample average can be written as an expectation: we have changed the averaging distribution to the empirical one. It is not an assertion that $\hat{P}_m$ equals the unknown true distribution.

For the sequence (A, A, A, B, C), the empirical distribution over (A, B, C) is $(3/5, 1/5, 1/5)$, not $(1/3, 1/3, 1/3)$. If we instead regard the five observation *indices* as equally likely, each has mass $1/5$; mapping indices to their values combines the repeated A outcomes. Keeping those spaces distinct avoids many empirical-entropy mistakes.

1.2.11 Every step of the MLE–cross-entropy–KL equivalence

For a discrete i.i.d. model with probabilities positive on observed outcomes,

$$\begin{aligned}
\arg\max_\theta Q_\theta(\mathcal{D}) &= \arg\max_\theta \prod_{i=1}^m Q_\theta(x^{(i)}) && \text{(independence)}\\
&= \arg\max_\theta \sum_{i=1}^m \log Q_\theta(x^{(i)}) && \text{(monotone log)}\\
&= \arg\max_\theta \frac{1}{m}\sum_{i=1}^m \log Q_\theta(x^{(i)}) && \text{(positive scaling)}\\
&= \arg\max_\theta \mathbb{E}_{\hat{P}_m} \log Q_\theta(X) && \text{(empirical mean)}\\
&= \arg\min_\theta \mathbb{E}_{\hat{P}_m}[-\log Q_\theta(X)] && \text{(negation)}\\
&= \arg\min_\theta H(\hat{P}_m, Q_\theta) && \text{(definition)}\\
&= \arg\min_\theta \{H(\hat{P}_m, Q_\theta) - H(\hat{P}_m)\} && \text{(constant in } \theta)\\
&= \arg\min_\theta D_{\mathrm{KL}}(\hat{P}_m \| Q_\theta). && (1.14)
\end{aligned}$$

These are equal sets of optimizers, not equal numerical objectives. In particular, the sum of log probabilities equals m times the empirical expectation, not the expectation itself. The factor disappears only because a positive constant does not change the argmax. Likewise, removing empirical entropy changes the objective value but not its optimizer or gradient.

Which way does the KL point? The empirical distribution comes first: it supplies the data being averaged. The model comes second: its probabilities are evaluated on those data. This is forward KL, $D_{\mathrm{KL}}(\hat{P}_m \| Q_\theta)$, not $D_{\mathrm{KL}}(Q_\theta \| \hat{P}_m)$. The order is determined by the likelihood calculation, not by a freely chosen convention.

Population interpretation. If the relevant expectations are finite, the population objective is

$$\mathbb{E}_{P_{\text{data}}}[-\log Q_\theta(X)] = H(P_{\text{data}}) + D_{\mathrm{KL}}(P_{\text{data}} \| Q_\theta).$$

Training substitutes a sample average for this unknown expectation. A law of large numbers gives convergence for a fixed θ under suitable assumptions; consistency of the *optimizer* requires additional conditions, such as identifiability and control of the objective over the family. Finite-sample maximum likelihood does not automatically find the true distribution. If the family is misspecified, even a population-optimal parameter can retain positive KL.

1.2.12 Worked MLEs: Bernoulli and categorical models

For a Bernoulli model $Q_\theta(1) = \theta$, let s of m labels be one. The likelihood of the observed ordered sequence is

$$\mathcal{L}(\theta) = \theta^s (1-\theta)^{m-s}.$$

If only the count is observed, its binomial probability includes $\binom{m}{s}$, but this factor is independent of θ and leaves the estimate unchanged. For $0 < s < m$,

$$\frac{d}{d\theta}\log \mathcal{L}(\theta) = \frac{s}{\theta} - \frac{m-s}{1-\theta} = 0 \quad \Longrightarrow \quad \hat{\theta} = \frac{s}{m}.$$

The second derivative is $-s/\theta^2 - (m-s)/(1-\theta)^2 < 0$, proving this is the unique interior maximum. Seven successes in ten trials therefore give 0.7. With all successes the maximum is at

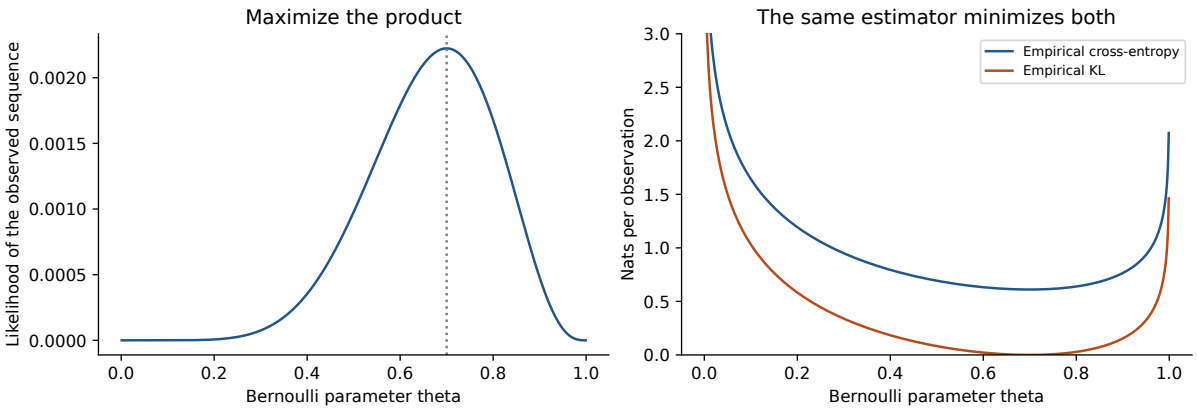


Figure 1.11: Seven successes and three failures. The product likelihood peaks at 0.7. Mean negative log-likelihood and empirical KL have the same minimizer; their vertical separation is the entropy of the empirical Bernoulli distribution.

$\theta = 1$ if the closed interval $[0, 1]$ is allowed; a finite-logit parameterization cannot reach it and only approaches the supremum. Parameter space matters.

For a K-class categorical model with counts n_k and probabilities q_k, maximize $\sum_k n_k \log q_k$ subject to $\sum_k q_k = 1$. When all counts are positive, the Lagrangian

$$\Lambda(q, \lambda) = \sum_k n_k \log q_k + \lambda \left(1 - \sum_k q_k\right)$$

gives $n_k/q_k = \lambda$. Summing $q_k = n_k/\lambda$ gives $\lambda = m$, so $q_k = n_k/m$. Concavity of the log objective makes this a global optimum. Zero-count classes obtain zero probability in the unrestricted closed simplex; this can produce infinite future log loss if a previously unseen class appears. Pseudocounts or priors can address that issue, but then the estimator is no longer this unregularized maximum-likelihood estimator. With a finite softmax parameterization, zero probabilities are approached only as limits.

1.2.13 Continuous data: densities are not point probabilities

A continuous measurement such as position has $P(X = x) = 0$ at every individual point, so a continuous-data likelihood uses a *density* $q_\theta(x)$, not point mass. For a small measurement bin of width Δ,

$$P_\theta(x \le X < x + \Delta) \approx q_\theta(x)\Delta.$$

With a fixed measurement scale, the product of Δ factors does not depend on θ, explaining why maximizing the density likelihood is meaningful. A density can exceed one, and a negative log density can be negative.

For densities p, q relative to the same base measure, define differential entropy and cross-entropy when the integrals exist:

$$h(P) = -\int p(x) \log p(x)\, dx, \qquad h(P, Q) = -\int p(x) \log q(x)\, dx.$$

Unlike discrete entropy, differential entropy can be negative and depends on units. For $U = aX + b$, $a \neq 0$, change of variables yields

$$p_U(u) = \frac{p_X((u - b)/a)}{|a|}, \qquad h(U) = h(X) + \log |a|.$$

Converting meters to centimeters adds $\log 100$ nats. Differential entropy by itself is therefore not an absolute number of bits needed to encode a real number exactly. Fine quantization gives, under appropriate regularity, $H(X_\Delta) \approx h(X) + \log(1/\Delta)$ in one dimension.

By contrast,

$$D_{\mathrm{KL}}(P\|Q) = \int p(x) \log \frac{p(x)}{q(x)}\, dx$$

is invariant under a common invertible smooth coordinate transformation: the Jacobian factors cancel inside the density ratio, and the outer measure changes consistently. It remains nonnegative, and it equals $h(P, Q) - h(P)$ when that subtraction is well defined.

A subtle limit of the empirical-KL shorthand

For continuous observations, the empirical measure is $\hat{P}_m = m^{-1} \sum_i \delta_{x^{(i)}}$, a collection of point masses. A smooth density model assigns zero probability to that finite set, whereas $\hat{P}_m$ assigns it probability one. Thus $D_{\mathrm{KL}}(\hat{P}_m\|Q_\theta) = +\infty$ in the literal measure-theoretic sense. The finite sample negative log-density is still a valid MLE objective, but it is not a finite KL between those two measures. Use the population-density KL interpretation, or explicitly discretize/smooth consistently. The discrete chain in (1.14) should not be transferred blindly to Dirac empirical measures and continuous densities.

This issue does not invalidate conditional classification: for each observed input, the predicted distribution over a finite set of labels is discrete. The per-example categorical cross-entropy remains well defined even when the input itself is continuous.

1.2.14 Interpreting the slide: approximating two modes by one Gaussian

Suppose a robot's position belief has two plausible locations because two corridors look similar. Use the explicit illustrative density

$$p(x) = \tfrac{1}{2}\mathcal{N}(x; -3, 0.7^2) + \tfrac{1}{2}\mathcal{N}(x; 3, 0.7^2).$$

We are forced to summarize it with a single Gaussian $q_{\mu,s}(x) = \mathcal{N}(x; \mu, s^2)$, $s > 0$. No member of this family can represent both narrow peaks faithfully. Which compromise does each KL choose?

Forward KL: derive the moment-matching solution. Minimizing $D_{\mathrm{KL}}(P\|Q_{\mu,s})$ is equivalent to minimizing $\mathbb{E}_P[-\log q_{\mu,s}(X)]$, because $h(P)$ is constant. The Gaussian negative log density gives

$$h(P, Q_{\mu,s}) = \frac{1}{2}\log(2\pi s^2) + \frac{\mathbb{E}_P[(X-\mu)^2]}{2s^2}.$$

Let $m_P = \mathbb{E}_P X$ and $v_P = \mathrm{Var}_P X$. Expanding $X - \mu = (X - m_P) + (m_P - \mu)$ and noting the cross term has mean zero gives

$$\mathbb{E}_P[(X-\mu)^2] = v_P + (m_P - \mu)^2.$$

For any s, the minimizing mean is $\mu = m_P$. Write $v = s^2$; then

$$\frac{d}{dv}\left[\frac{1}{2}\log(2\pi v) + \frac{v_P}{2v}\right] = \frac{v - v_P}{2v^2}.$$

It is negative below v_P and positive above, so the minimizing variance is $v = v_P$ when $v_P > 0$. For our symmetric mixture,

$$\boxed{\mu^* = 0, \qquad (s^*)^2 = 3^2 + 0.7^2 = 9.49, \qquad s^* \approx 3.0806.}$$

The broad fitted Gaussian covers both modes, even though it assigns appreciable density to the low-probability region between them. Ignoring either mode would make many P-drawn observations extremely surprising. That is what forward KL penalizes. The mean position zero is a statistical summary, not necessarily a physically plausible robot location.

Reverse KL: why a single mode can win. The reverse objective is

$$D_{\mathrm{KL}}(Q_{\mu,s}\|P) = \mathbb{E}_Q[\log q_{\mu,s}(X) - \log p(X)] = -h(Q_{\mu,s}) - \mathbb{E}_Q \log p(X).$$

Now the fitted model supplies the samples. A broad Q generates many samples between the two peaks, where $p(x)$ is tiny, producing a large penalty. A narrow Q centered near one peak rarely visits the valley or the other peak. Missing the other mode is less directly penalized because expectations are under Q. The $-h(Q)$ term matters too: it discourages collapsing a continuous Gaussian to an infinitesimally narrow spike. Reverse KL is not merely "maximize p at the model mean."

Near the left mode, if the peaks are well separated, $p(x) \approx \frac{1}{2}\mathcal{N}(x; -3, 0.7^2)$. Choosing Q close to that component makes $\log(q/p) \approx \log 2$, so its reverse KL is roughly 0.6931 nats. The companion script searches Gaussian mean and scale on a grid, refines the best grid neighborhood, and evaluates expectations by Gauss–Hermite quadrature. It finds a fit near $(\mu, s) = (-3, 0.7)$ with reverse KL about 0.69312. By symmetry a corresponding fit near $(3, 0.7)$ has the same value. This numerical search supports the displayed example; it is not a theorem that reverse KL always selects exactly one mode or a proof of global optimality for every mixture.

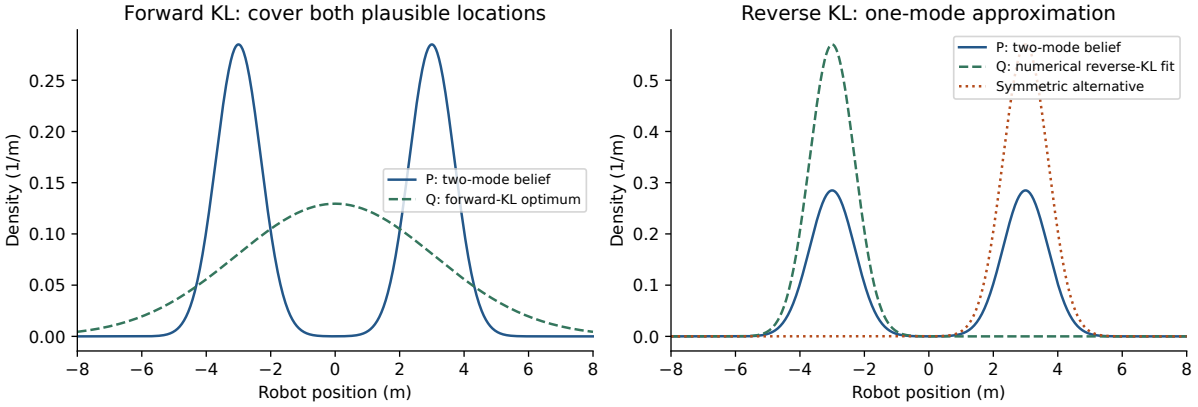


Figure 1.12: A constructed counterpart of the slide's qualitative example, with explicit distributions. Forward KL has an analytically derived Gaussian optimum. Reverse KL has two symmetric one-mode candidates found numerically. The mismatch is unavoidable because a single Gaussian cannot express this two-mode belief.

When the slogan fails. "Forward KL covers modes; reverse KL seeks modes" is a useful tendency for restricted approximating families and well-separated targets. If the family contains P, both directions can achieve the exact match. Overlapping modes, unequal mixture weights, or a richer family can change the compromise. Neither direction is universally better. For localization, throwing away a plausible mode can hide ambiguity, while a broad Gaussian can place probability in impossible corridors. A multimodal representation may be preferable to either single-Gaussian approximation.

Why maximum likelihood is the left-hand case. MLE averages $-\log q_\theta(x)$ over *observed data*. It asks whether the model makes the data plausible, giving forward KL in the population/discrete interpretation. It does not sample from Q_θ and ask whether those samples are

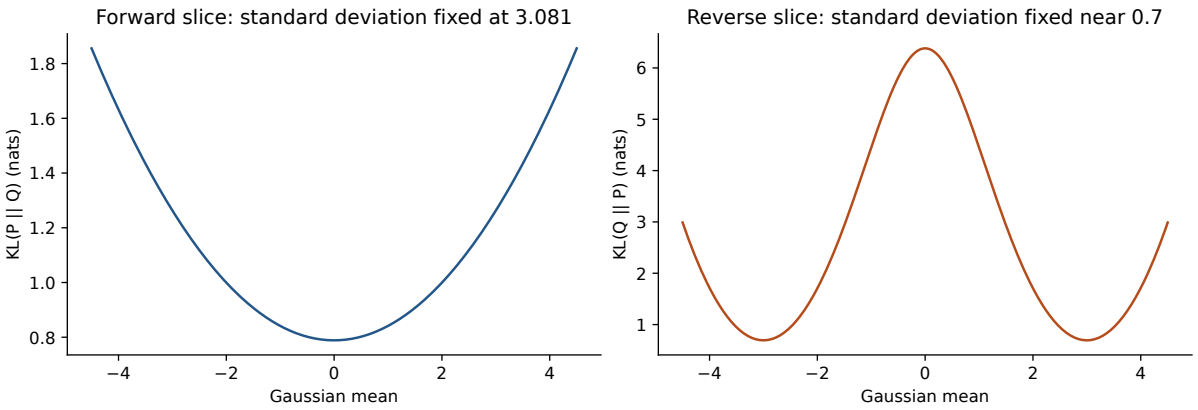


Figure 1.13: One-dimensional slices through two different objectives. Each panel holds the standard deviation at its displayed fitted value and varies the mean. The forward slice favors the midpoint; the reverse slice favors either peak. These slices are not the full two-parameter optimization landscapes.

plausible under P. Reverse-KL objectives arise in other settings, but do not follow from the likelihood derivation just given.

1.2.15 Transfer question 1: why track KL rather than cross-entropy?

Because KL removes the uncertainty floor. A perfect model of a fair binary source still has cross-entropy $\log 2 \approx 0.6931$ nats. A perfect model of a source with probabilities $(0.95, 0.05)$ has cross-entropy about 0.1985 nats. Both have KL zero. A larger cross-entropy does not necessarily mean the model is further from its source; the source may simply be less predictable.

For a fixed known P, KL states the excess expected loss relative to the unrestricted optimum $Q = P$. For example, if $H(P) = 0.60$ and $H(P, Q) = 0.65$, the model pays 0.05 extra nats per observation. Cross-entropy alone would not reveal that gap unless we knew the entropy floor.

Three qualifications make this useful in practice. First, with P fixed, subtracting $H(P)$ leaves gradients and optimizer locations unchanged. Computing KL does not improve optimization by itself. Second, if the model family cannot represent P, its best attainable KL may be positive. The actual optimization gap within that family is

$$D_{\mathrm{KL}}(P\|Q_\theta) - \inf_\vartheta D_{\mathrm{KL}}(P\|Q_\vartheta),$$

not simply $D_{\mathrm{KL}}(P\|Q_\theta)$. Third, true P and its entropy are usually unknown. Held-out samples let us estimate cross-entropy, while estimating entropy accurately can be difficult, especially in high dimensions.

If labels are represented by one-hot targets, the target entropy for each example is zero, so per-example cross-entropy and KL to that point mass are numerically the same. That does not prove the true conditional label entropy is zero. With noisy or ambiguous labels, the real uncertainty can remain positive. Known simulator distributions, repeated outcomes at the same finite context, or soft target distributions are settings in which the entropy subtraction can be directly informative.

1.2.16 Transfer question 2: derive binary cross-entropy as a special case

For an input x_n, the model's distribution over labels $0, 1$ is

$$Q_n = (1 - \hat{y}_n, \hat{y}_n), \qquad \hat{y}_n = p_w(Y = 1 \mid x_n).$$

The observed label $y_n \in \{0, 1\}$ defines a target distribution concentrated on that outcome:

$$P_n = (1 - y_n, y_n).$$

Substituting these two probability vectors into the general finite-outcome cross-entropy, with the sum over *labels*, gives

$$\begin{aligned} H(P_n, Q_n) &= -\sum_{k \in \{0,1\}} P_n(k) \log Q_n(k) \\ &= -(1 - y_n) \log(1 - \hat{y}_n) - y_n \log \hat{y}_n. \end{aligned}$$

Averaging over the N observations gives exactly the lecture formula:

$$\boxed{L(w) = -\frac{1}{N} \sum_{n=1}^{N} \left[y_n \log \hat{y}_n + (1 - y_n) \log(1 - \hat{y}_n) \right].}$$

There are two different averages: the inner sum is over possible labels at a fixed input; the outer sum is over dataset examples. The labels and predictions across all examples are not themselves one normalized probability vector.

An equivalent joint construction avoids ambiguity: put mass $1/N$ on each example index n, and define $P(n, k) = N^{-1}\mathbf{1}\{k = y_n\}$ and $Q_w(n, k) = N^{-1}Q_n(k)$. Then

$$H(P) = \log N, \quad H(P, Q_w) = \log N + L(w), \quad D_{\mathrm{KL}}(P \| Q_w) = L(w).$$

This index-based joint space is not the same as grouping repeated inputs into an empirical conditional label distribution, whose entropy may be positive. Both constructions produce the same averaged conditional log loss; they assign entropy to different random variables.

Soft targets and more than two classes. If a target supplies a probability $r_n \in [0, 1]$ rather than a hard label, replace y_n by r_n. The same formula becomes the cross-entropy between two Bernoulli distributions. Its KL equals that BCE minus the target entropy. For K classes, target probabilities r_{nk} and predictions q_{nk} give

$$L = -\frac{1}{N} \sum_{n} \sum_{k=1}^{K} r_{nk} \log q_{nk}.$$

With one-hot targets this reduces to the negative log probability of the correct class. Label smoothing changes the target distribution, hence also its entropy and the optimum; it is not merely a different algebraic form of the original hard-label objective.

1.2.17 Worked semantic-perception example: separate uncertainty from error

Consider a robot semantic classifier with labels free space, person, and cart. For a transparent teaching example, the input has only two discrete contexts: open floor and occluded view. In 100 observations of each context, the label counts are

Context	Free	Person	Cart
Open floor	80	15	5
Occluded view	10	20	70

The empirical conditional distributions are $\hat{P}_A = (0.80, 0.15, 0.05)$ and $\hat{P}_B = (0.10, 0.20, 0.70)$. Suppose a model predicts $Q_A = (0.70, 0.20, 0.10)$ and $Q_B = (0.20, 0.30, 0.50)$. Both contexts have weight $1/2$ in this dataset. Directly from the counts,

$$\mathrm{NLL}_{\mathrm{mean}} = -\frac{1}{200} \sum_{c \in \{A,B\}} \sum_{k} n_{ck} \log Q_c(k) \approx 0.764413 \text{ nats.}$$

The empirical conditional entropy and mismatch are

$$\hat{H}(Y \mid X) = \tfrac{1}{2} H(\hat{P}_A) + \tfrac{1}{2} H(\hat{P}_B) \approx 0.707344,$$
$$\hat{D} = \tfrac{1}{2} D_{\mathrm{KL}}(\hat{P}_A \| Q_A) + \tfrac{1}{2} D_{\mathrm{KL}}(\hat{P}_B \| Q_B) \approx 0.057069.$$

They add to 0.764413, as predicted by (1.13). An unrestricted categorical model per context would fit the empirical frequencies and reduce the empirical KL to zero, but its cross-entropy would remain 0.707344. The remaining label variation cannot be predicted from these two context categories in this empirical model.

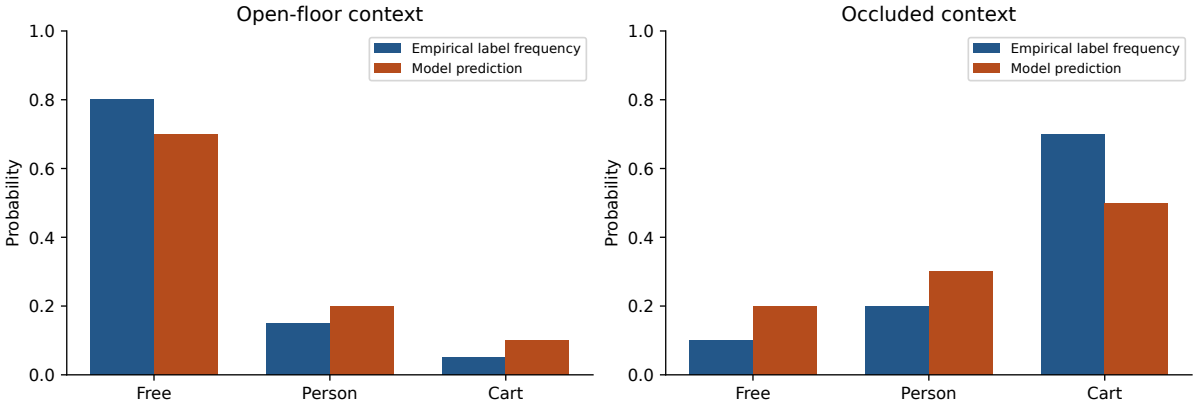


Figure 1.14: The robot example's target frequencies and model probabilities. The occluded context underpredicts carts; the resulting conditional KL measures that mismatch separately from the label variability within each context.

This is an in-sample decomposition of constructed counts, not an estimate of irreducible uncertainty on all future robot observations. Richer inputs might resolve ambiguity; finite frequencies are noisy; context proportions can shift. For a real study, fit models on training data, compare predictive log loss on held-out objects or sessions, and report how any entropy estimate was obtained. Do not estimate an entropy floor from the same overfit model and then call it known ground truth.

1.2.18 Implement the definitions without hiding their edge cases

The companion uses NumPy for the definitions and numerical experiments. A stable finite-distribution KL calculation must ignore zero P terms while returning infinity when positive P meets zero Q:

```
def kl(p, q):
    p = np.asarray(p, dtype=float)
    q = np.asarray(q, dtype=float)
    # Validate matching shapes, nonnegativity and sums of one first.
    active = p > 0
    if np.any(q[active] == 0):
        return float("inf")
    return np.sum(p[active] *
                  (np.log(p[active]) - np.log(q[active])))
```

The full script validates probability vectors and supports bits or nats. Using a mask before evaluating logarithms avoids invalid inactive terms. Adding an arbitrary epsilon to every probability changes the distributions and may destroy normalization. If smoothing is intended, define and normalize the smoothed model explicitly. For network outputs, compute log probabilities from logits using stable log-sum-exp; avoid taking logs after probabilities have rounded to zero.

In the mixture example, use

$$\log p(x) = \operatorname{logaddexp}\big(\log \mathcal{N}(x; -3, 0.7^2), \log \mathcal{N}(x; 3, 0.7^2)\big) - \log 2.$$

Computing each tiny density and adding them first can underflow in the tails. For reverse KL, write $X = \mu + sZ$, $Z \sim \mathcal{N}(0, 1)$, and integrate over Z. Gauss–Hermite quadrature approximates this expectation without truncating the real line to a plotting window. The script checks the selected candidate with both 120 and 240 quadrature nodes; their reverse-KL estimates differ by less than 10^{-8} for this example. Numerical integration and grid search remain approximations, not symbolic proofs of an optimum.

Reproduce and inspect. From the repository root:

```
python deep_learning/code/information_1_2.py
make dl
```

Seven vector figures and a numerical summary are written under `deep_learning/figures/information_1_2/`. Committed figures allow the notes to compile without running Python. The script checks the entropy decomposition, KL support cases, empirical-NLL equivalence, the BCE special case, and coordinate invariance for a Gaussian KL. It uses no machine-learning library and labels all examples as constructed data.

1.2.19 Reasoning exercises

Try these before the worked answers. Explain what distribution supplies the expectation in every expression, rather than starting with algebra alone.

Exercise 1.6 (An exact code)**.** For $P = (1/2, 1/4, 1/8, 1/8)$, use the three codebooks above to verify each expected length. How many extra ideal bits does the uniform code use over 1,000 i.i.d. events on average? Does this say every sequence uses exactly that many extra bits?

Exercise 1.7 (Asymmetry and impossible events)**.** Evaluate both KL directions for $P = (1, 0)$ and $Q = (1/2, 1/2)$ in bits. Which expectation encounters an impossible event, and under which model?

Exercise 1.8 (MLE without a remembered formula)**.** For a categorical dataset with counts $(12, 6, 2)$, derive its unrestricted MLE using a Lagrange multiplier. What changes if the last count is zero?

Exercise 1.9 (Two sources, the same reported loss)**.** Two models each have cross-entropy 0.7 nats. Their source entropies are 0.69 and 0.2. Which model is closer in forward KL to its own source? Can you infer which has higher classification accuracy?

Exercise 1.10 (Moments and modes)**.** For $P = \frac{1}{2}\mathcal{N}(-a, s_0^2) + \frac{1}{2}\mathcal{N}(a, s_0^2)$, derive the forward-KL Gaussian mean and variance. Explain what happens as $a \to 0$ and why this limits a universal "reverse KL chooses one peak" slogan.

Exercise 1.11 (One-hot targets are not a population claim)**.** A repeated input produces label one in 70% of trials. What probability minimizes expected BCE? Why can every observed one-hot target have zero entropy while the conditional source entropy is positive?

1.2.20 Worked answers and checks on interpretation

1. Coding. The averages are 1.75, 2, and 2.625 bits per event. The uniform code costs $0.25 \times 1000 = 250$ extra bits in expectation. Individual sequences have fluctuating outcome counts, so the realized difference need not equal 250. The statement concerns an average under the source distribution.

2. Support. $D_{\mathrm{KL}}(P\|Q) = \log_2(1/(1/2)) = 1$ bit. The reverse direction is infinite: Q samples the second event with probability $1/2$, but P assigns it zero probability. In the forward direction that event is never sampled.

3. Categorical MLE. The stationary equations give $q_k = n_k/\lambda$ and $\lambda = 20$, hence $(0.6, 0.3, 0.1)$. If the dataset instead has counts $(12, 6, 0)$, its total is 18 and the closed-simplex optimum is $(2/3, 1/3, 0)$. A finite softmax can approach but not reach this boundary. A future third-class observation would have infinite loss under the unsmoothed boundary solution.

4. Excess loss. The KL values are 0.01 and 0.5 nats. The first is much closer in this directed log-loss sense, even though raw cross-entropies agree. Classification accuracy cannot be inferred from these quantities alone: it discards confidence and also depends on a decision rule and the source distribution.

5. Gaussian approximation. The mean is zero by symmetry. The second moment is $a^2 + s_0^2$, so the variance is also $a^2 + s_0^2$. As $a \to 0$, the mixture becomes one Gaussian and both KL directions can achieve zero with that same Gaussian. The two-mode behavior is about approximating a separated multimodal target with a restricted family.

6. Label noise. The expected loss is $-0.7 \log p - 0.3 \log(1 - p)$ and is minimized at $p = 0.7$. Each realized label defines a point mass with entropy zero, but before the outcome is known the conditional label distribution is Bernoulli(0.7), with entropy about 0.610864 nats. An observed target and the process generating successive targets are different distributions.

A way to reconstruct the whole section

Read the expectation first: it tells you who supplies the outcomes. Then read the logarithm: it tells you whose probabilities judge them. If both are P, you measure entropy. If P supplies outcomes and Q scores them, you measure cross-entropy. Their excess is forward KL. Replacing the unknown source expectation by an empirical average gives the negative log-likelihood used for learning. Always identify the outcome space, conditioning, support, and log unit before manipulating the formula.

Relation to the lecture and further reading. This section expands the supplied entropy, KL, and maximum-likelihood slides, including both transfer questions and the qualitative two-mode diagram. The new diagrams use explicitly specified distributions and reproducible calculations. For the lecture's textbook context, see Goodfellow et al. [2], Chapter 3, Section 3.13 and Figure 3.6, and Chapter 5, Section 5.5 (especially the maximum-likelihood equivalences). The finite-code qualifications and the continuous empirical-measure distinction above specify the assumptions needed to interpret the concise slide statements carefully.

1.3 Logistic Regression as a Neural Network: The Perceptron

1.3.1 One model, two ways of seeing it

Logistic regression asks for the probability of a binary outcome given measured features. A neural-network diagram exposes the computations that produce that probability. It does not, by itself, introduce a different statistical model. For $\mathbf{x} \in \mathbb{R}^d$, weights $\mathbf{w} \in \mathbb{R}^d$, and bias $b \in \mathbb{R}$,

$$z = \mathbf{w}^\mathsf{T}\mathbf{x} + b = \sum_{j=1}^{d} w_j x_j + b, \tag{1.15}$$

$$p = \sigma(z) = \frac{1}{1 + e^{-z}}. \tag{1.16}$$

The score z is called the *pre-activation* or *logit*; p is the activation and the model's estimate of $P(Y = 1 \mid \mathbf{x})$. The bias is the same quantity denoted w_0 in the preceding sections. The first operation is technically *affine*, since the bias allows a nonzero output at zero input.

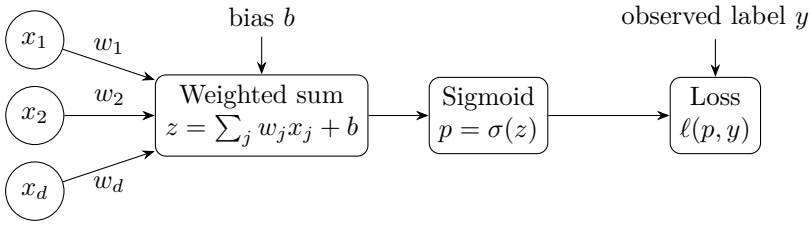


Figure 1.15: A sigmoid neuron and its training loss. The label enters the loss computation, not the prediction. Intermediate inputs are omitted when $d > 3$.

Why can z be interpreted as accumulated evidence? From the sigmoid,

$$1 - p = \frac{e^{-z}}{1 + e^{-z}}, \qquad \frac{p}{1 - p} = e^{z}, \qquad \log \frac{p}{1 - p} = z = b + \sum_j w_j x_j.$$

Thus an increase Δ in feature x_j, with all other features fixed, changes log-odds by $w_j\Delta$ and multiplies odds by $e^{w_j\Delta}$. The sigmoid translates this additive evidence into a number between zero and one. This is a modeling assumption about conditional log-odds, not a theorem that every classification problem has this structure.

What does "perceptron" mean here?

The classical perceptron uses a hard threshold, $\hat{y} = \mathbf{1}\{z \geq 0\}$, and its classical learning rule differs from logistic regression training. The lecture diagram uses the term for a sigmoid neuron. Logistic regression is exactly this sigmoid neuron trained with binary cross-entropy. Hard threshold outputs are not suitable probability estimates for this loss: a confidently wrong prediction gives infinite negative log-likelihood, and the threshold is not differentiable at zero and has zero derivative elsewhere.

At the usual probability threshold $1/2$, sigmoid monotonicity implies

$$p \geq \tfrac{1}{2} \iff z \geq 0.$$

The decision boundary remains $\mathbf{w}^\mathsf{T}\mathbf{x} + b = 0$. A nonlinear output activation therefore does not automatically create a nonlinear decision boundary in the original input features. Hidden nonlinear features, introduced in later sections, can change that geometry.

1.3.2 A complete numerical forward pass

Consider an illustrative grasp-success model using standardized contact-quality and pose-error features. Set

$$\mathbf{x} = \begin{bmatrix} 1.2 \\ 0.5 \end{bmatrix}, \qquad \mathbf{w} = \begin{bmatrix} 2 \\ -1.5 \end{bmatrix}, \qquad b = -0.4.$$

The signs encode the model's assumptions: better contact increases predicted success, while larger pose error reduces it. These are illustrative weights, not a validated robot controller. Compute

$$z = 2(1.2) - 1.5(0.5) - 0.4 = 1.25, \qquad p = \sigma(1.25) \approx 0.7773.$$

If the grasp succeeds ($y = 1$), its loss is $-\log p \approx 0.2519$ nats. If it fails ($y = 0$), the same prediction incurs $-\log(1-p) \approx 1.5019$ nats. The forward prediction is identical in both cases; the observed outcome determines how it is scored.

1.3.3 Computational complexity: count the forward-pass work

Assume a dense dataset with N examples and $d \geq 1$ features, and fixed-precision arithmetic where a scalar arithmetic operation, exponential, or logarithm has constant cost in N and d. For one example, computing z uses d multiplications, $d-1$ additions to sum the products, and one bias addition. This is $\Theta(d)$ work. The sigmoid then operates on one scalar, so it costs $\Theta(1)$. Binary cross-entropy also uses only a fixed number of scalar operations:

$$\ell(p, y) = -y \log p - (1 - y) \log(1 - p).$$

Repeating for all examples and averaging gives

$$J(\mathbf{w}, b) = \frac{1}{N} \sum_{n=1}^{N} \ell(p_n, y_n).$$

Operation	One example	Entire dataset
Weighted sum and bias	$\Theta(d)$	$\Theta(Nd)$
Sigmoid activation	$\Theta(1)$	$\Theta(N)$
Per-example cross-entropy	$\Theta(1)$	$\Theta(N)$
Sum and average losses	—	$\Theta(N)$

Consequently,

$$\boxed{\Theta(Nd + N) = \Theta(Nd), \quad \text{hence } O(Nd).}$$

The weighted sums dominate because every example must combine all d input features. If both N and d double, the dominant work increases by a factor of four. For example, $N = 10{,}000$ and $d = 100$ require 10^6 weighted feature products; doubling both requires 4×10^6.

1.3.4 The same argument as a matrix multiplication

Store examples as rows:

$$X = \begin{bmatrix} \mathbf{x}_1^\mathsf{T} \\ \vdots \\ \mathbf{x}_N^\mathsf{T} \end{bmatrix} \in \mathbb{R}^{N \times d}, \quad \mathbf{w} \in \mathbb{R}^d, \quad \mathbf{z} = X\mathbf{w} + b\mathbf{1} \in \mathbb{R}^N.$$

Matrix–vector multiplication computes N dot products, each of length d: $\Theta(Nd)$. Applying sigmoid elementwise produces $\mathbf{p} \in \mathbb{R}^N$ in $\Theta(N)$ time, followed by $\Theta(N)$ loss evaluation and reduction. Vectorization can improve execution speed through optimized numerical kernels; it does not remove the Nd arithmetic dependence.

What exactly is being counted?

Starting from inputs and weights, computing the full loss costs $O(Nd)$. If probabilities are already available, computing cross-entropy alone costs $O(N)$. This is the cost of one evaluation, not the whole optimization process. A mini-batch of B examples costs $O(Bd)$. Sparse matrix–vector multiplication can reduce the full forward-pass cost to $O(\mathrm{nnz}(X) + N)$, where $\mathrm{nnz}(X)$ counts stored nonzero entries.

1.3.5 A stable NumPy implementation

Directly taking logarithms of computed probabilities can fail when floating-point sigmoid outputs round to zero or one. Substitute $p = \sigma(z)$ into the loss instead:

$$\log p = z - \log(1 + e^z), \tag{1.17}$$

$$\log(1 - p) = -\log(1 + e^z), \tag{1.18}$$

$$\ell(z, y) = \log(1 + e^z) - yz. \tag{1.19}$$

For binary y, an equivalent expression is

$$\ell(z, y) = \log(1 + e^{(1-2y)z}).$$

Evaluate it using `np.logaddexp`, which avoids forming an overflowing exponential. This form also avoids subtracting two large nearly equal terms for a confident correct prediction.

```
import numpy as np

def binary_neuron_loss(X, y, w, b):
    # X: (N, d); y: (N,), binary; w: (d,); b: scalar
    z = X @ w + b                          # O(Nd)
    signed_logits = (1.0 - 2.0*y) * z   # O(N)
    losses = np.logaddexp(0.0, signed_logits)  # O(N)
    return losses.mean()                # O(N)

X = np.array([[1.2, 0.5]])
w = np.array([2.0, -1.5])
assert np.isclose(binary_neuron_loss(X, np.array([1.]),
                                     w, -0.4), 0.2519290813)
assert np.isclose(binary_neuron_loss(X, np.array([0.]),
                                     w, -0.4), 1.5019290813)
```

Materializing logits and losses uses $O(N)$ additional memory beyond the dataset and parameters. Processing one example at a time permits a running sum with constant extra workspace beyond the current input and weights; mini-batches trade memory against throughput.

1.3.6 Connection to learning and the next layer

The chain rule gives a particularly simple error signal:

$$\frac{\partial \ell}{\partial p} = \frac{p - y}{p(1 - p)}, \qquad \frac{\partial p}{\partial z} = p(1 - p), \qquad \frac{\partial \ell}{\partial z} = p - y.$$

These identities hold for finite logits, hence $0 < p < 1$. Since $\partial z/\partial w_j = x_j$ and $\partial z/\partial b = 1$,

$$\nabla_{\mathbf{w}} J = \frac{1}{N} X^{\mathsf{T}}(\mathbf{p} - \mathbf{y}), \qquad \frac{\partial J}{\partial b} = \frac{1}{N} \sum_n (p_n - y_n).$$

The gradient also costs $O(Nd)$. Thus a full-batch gradient step, including its forward pass, has the same asymptotic order; T such steps cost $O(TNd)$. Convergence determines the required T and is a separate question.

A single sigmoid neuron learns one affine score. Placing several neurons in a hidden layer lets the model construct several features and combine them. If there are h hidden neurons, their dense forward computation is $XW + \mathbf{1}\mathbf{b}^\mathsf{T}$ with $W \in \mathbb{R}^{d\times h}$, costing $O(Ndh)$. This explains both the computational growth and the next conceptual step toward a multilayer perceptron.

1.4 Multilayer Perceptrons

1.5 Matrix Dimensions

1.6 Other Activation Functions and Loss Functions

1.7 Representational Power of MLPs

1.8 Summary, Further Reading, References

Chapter 2

MLPs and Backpropagation

2.1 Recap of Multilayer Perceptrons

2.2 Recap of Chain Rule of Calculus

2.3 Calculating Gradients with Backpropagation

2.4 Backprop for a Linear Layer in Matrix Form

2.5 Full Matrix Form of Backpropagation Equations

2.6 Summary, Further Reading, References

Bibliography

[1] Christopher M. Bishop. *Pattern Recognition and Machine Learning*. Springer, 2006.

[2] Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep Learning*. MIT Press, 2016. URL https://www.deeplearningbook.org.